

# Can Agentic AI Settle a 20-Year-Old Description Logic Problem?

On the Decidability of  $\mathcal{ALCI}_{\text{RCC5}}$ , a Decidable Fragment,  
and a Certified Lean Formalization Reducing It to One Keystone

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with AI assistants:

Claude Opus 4.6, Opus 4.7, Opus 4.8, Fable 5, Opus 5 (Anthropic)  
GPT-5.4, GPT-5.4 Pro, GPT-5.5, GPT-5.6 Pro, GPT-5.6 Sol (OpenAI)

August 2026

## Abstract

Whether concept satisfiability in  $\mathcal{ALCI}_{\text{RCC5}}$ —the description logic  $\mathcal{ALCI}$  with the RCC5 base relations as roles under *abstract composition-table* semantics—is decidable has been open since Wessel’s 2002/2003 work. This paper reports a sustained, adversarially-reviewed attack on the problem, conducted by AI assistants under human direction, and machine-checked in part: the soundness pipeline is kernel-checked in Lean 4, the keystone is not. We do not settle the question. What we establish is: (i) a *conditional decidability theorem*—if the “live width” of models is bounded by a computable function of the concept (property F6), then satisfiability is decidable—whose soundness direction is *machine-checked in Lean 4*; (ii) that the remaining completeness problem of *this* finite-certificate architecture compresses into that single property F6—the *same horizontal-coordination obstruction* that appears in the known grid-style attempts to prove the problem undecidable; (iii) an RCC5 normal form (now machine-certified in *both* directions) and an unconditionally decidable fragment (the  $\forall\text{PO}$ -free concepts), whose decision procedure is itself *machine-checked* — on raw input and under concrete set semantics, with fragment membership its only hypothesis — and has survived three independent cold reviews; it is a decidability theorem, not a practical procedure, the search space being doubly exponential; (iv) a *working, empirically-validated prototype reasoner*—a cover-tree tableau that reproduces, 23 years on, the GIS concept taxonomy of the original report, cross-validated on 911 concepts with zero mismatches and never once contradicted by an independent oracle or a hand analysis; and (v) an honest map of the routes tried, the dead ends, and the missing hard pieces. We also report, candidly, on the methodology—adversarial “cold review,” machine verification, and a human-in-the-loop workflow—and on what frontier AI could and could not do.

**Keywords:** description logics;  $\mathcal{ALCI}$ ; RCC5; region connection calculus; qualitative spatial reasoning; decidability; concept satisfiability; patchwork property; interactive theorem proving (Lean); AI-assisted mathematics.

**ACM CCS:** Theory of computation → Description logics; Automated reasoning; Logic and verification. Computing methodologies → Spatial and physical reasoning.

**MSC 2020:** 03B70 (logic in computer science); 68T27 (logic in artificial intelligence); 68T30 (knowledge representation); 68V15 (theorem proving).

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## 1 Introduction

### 1.1 AI disclaimer and authorship

The mathematical content of this paper was produced by AI assistants (Anthropic’s Claude Opus 4.6, 4.7 and 4.8, Claude Fable 5 and Claude Opus 5; and OpenAI’s GPT-5.4, GPT-5.4 Pro, GPT-5.5, GPT-5.6 Pro and GPT-5.6 Sol), prompted, steered, and reviewed by Wessel, who posed

the problem, supplied the central intuitions and directions, orchestrated the review–repair cycles, and takes responsibility for the framing. **The results are unverified by human domain experts and should be read as AI-generated conjectures supported by machine checking and computational evidence—not as established theorems**, with the explicit exceptions of the parts stated to be kernel-checked in Lean 4 (the soundness pipeline, the abstraction’s faithfulness, the RCC5 normal form, and—since 2026-08-29—the decision procedure for the  $\forall\text{PO}$ -free fragment), which are machine-verified. We are deliberate about this labelling throughout, and it is not uniform: the standing one-line status for the *full logic* is *strongly supported, not certified*, whereas the  $\forall\text{PO}$ -free fragment is *machine-certified*, its formalization’s adequacy having been examined by three independent cold reviews (§7.3). What remains at risk there is what remains at risk in any formalization—that a definition misrepresents the intended object—which is why those reviews targeted the definitions, and why the concrete-semantics question was converted into a theorem rather than left to reviewer confidence. We regard the AI systems as having made contributions of a kind that, in a human collaborator, would warrant co-authorship, and we discuss this in §9.

## 1.2 A motivating example: what the logic does

Before the difficulties, a concrete picture of the object. In  $\mathcal{ALCZ}_{\text{RCC5}}$  the five RCC5 relations serve as *roles*: a concept may quantify over the regions a given region is part of, overlaps, or is disjoint from. A small GIS vocabulary (from Wessel’s original report [9]) shows what this buys. Define

$$\text{CityAtRiver} \equiv \text{City} \sqcap \exists \text{PO}.\text{River}, \quad \text{GermanCity} \equiv \text{City} \sqcap \forall \text{PP} . (\neg \text{Country} \sqcup \text{Germany})$$

—“a city overlapping a river” and “a city lying in no country but Germany”—and *assert* only a few primitive facts:

$$\begin{aligned} \text{Hamburg} &\sqsubseteq \text{City} \sqcap \exists \text{PO}.\text{Alster}, \\ \text{Alster} &\sqsubseteq \text{River} \sqcap \exists \text{PP}.\text{Germany} \sqcap \forall \text{PO}.\neg \text{Country} \sqcap \forall \text{PP} . (\neg \text{Country} \sqcup \text{Germany}). \end{aligned}$$

Nothing here *states* that Hamburg is a German city or a city at a river; a reasoner *infers* both. **CityAtRiver** is immediate—Hamburg overlaps the Alster and the Alster is a river, so Hamburg overlaps a river. **GermanCity** is subtler and, tellingly for what follows, runs through the *composition table*: for any country  $c$  with **Hamburg**  $\text{PP } c$ , the path **Alster**  $\text{PO Hamburg PP } c$  forces **Alster**  $R c$  with  $R \in \text{comp}(\text{PO}, \text{PP}) = \{\text{PO}, \text{PP}\}$  (§3); the Alster overlaps no country, which excludes  $\text{PO}$ , so **Alster**  $\text{PP } c$ ; and the Alster’s only country-superpart is Germany, so  $c = \text{Germany}$ . Hence Hamburg is a German city.

Two things survive into the rest of the paper. First, this is the *reasoning service*—classification—and it reduces, as all such services do, to one decision problem: *concept satisfiability* ( $C \sqsubseteq D$  iff  $C \sqcap \neg D$  is unsatisfiable), whose decidability is exactly what is open. Second, even this benign inference already leans on the RCC5 composition table; the whole difficulty is that the *same* table is, on other inputs, too loose to force what one would need (§1.3). The full taxonomy this example lives in is the one our reasoner recomputes 23 years on (§8, Figures 4–5); how the tableau actually derives the **GermanCity** inference is spelled out there.

## 1.3 Why the problem is hard, in one page

RCC5 relates two regions by exactly one of five relations: equal (EQ), proper part (PP), proper superpart (PPI), partial overlap (PO), or discreteness/disjointness (DR). Under *abstract composition-table* semantics a model is not a set of actual regions but a set with a total labelling of ordered pairs

by these relations, required only to be converse-closed and *composition-closed*: if  $x R y$  and  $y S z$  then  $x T z$  for some  $T$  the composition table allows.

The five relations split into two axes that behave completely differently.

- **The vertical axis** (PP, PPI: nesting) is *tame*. Composition can *force* vertical structure—the only four “determining” compositions (those returning a single relation) all run through PP or PPI—but an unbounded vertical chain is captured by a finite *generator*: a self-loop that *unfolds* into infinitely many distinct points (never an actual PP-cycle, which strong EQ forbids), just as regular structures capture infinite paths in modal logic. Infinite depth is not the problem.
- **The horizontal axis** (DR, PO: side-by-side) is *wild*. No *single* composition step pins a horizontal relation to one value (PO is never a singleton table entry)—though as constraints accumulate, their *intersection* sometimes can. What composition still cannot do is force two occurrences *together*; it can only force them *apart* (a shared horizontal anchor excludes EQ). So horizontal “crowds” of pairwise-distinct occurrences are the danger.

A finite decision procedure must record, at each interface of a model, the genuine branching it cannot reconstruct—the *live width*. Many forced edges are *shadows* (implied by composition from edges already present) and cost nothing; live width counts only the rest. The whole decidability question reduces to one property:

**(F6)** Is the live width of models bounded by a computable function of the input concept?

If yes, the certificate space is finite and satisfiability is decidable. And here is the knife’s edge: the looseness F6 must tame is the *same horizontal-coordination obstruction* that has blocked every known attempt to prove the problem *undecidable*. To force undecidability one would encode a rigid coordinate grid, which needs *clean, reusable coincidence*—a horizontal addressing scheme reused across the grid. Individual horizontal edges *can* be pinned to a single value by the joint closure of several constraints (Remark 2), but no concept is known to make such coincidence *reusable* rather than one-off (the “coincidence obstruction” Wessel identified in 2002/2003). Turn that looseness into a computable bound and one gets decidability by *this* route; circumvent it to reuse coincidence and force a grid and one gets undecidability. The two open directions face the same obstruction from two sides—a *shared* wall, not a theorem that F6’s failure would by itself *entail* undecidability (§10)—which is why neither has been settled in over twenty years.

## 1.4 Scope, contributions, and structure

This is both a technical report and a discussion piece. It is, specifically, an *overview and status map*: it states the results and their verification level but leaves the full proofs to the cited work packages and formal artifacts. Its contributions are:

1. A *conditional decidability theorem* ( $F6 \Rightarrow \text{decidable}$ ) whose soundness half is machine-checked in Lean 4 (§7.6), and the reduction of the whole problem to F6.
2. The identification of F6 as the shared *obstruction* facing both the present certificate route and the known grid-style undecidability attempts (§1.3, §10).
3. An RCC5 *normal form* (both directions machine-certified) and an unconditionally decidable  $\forall\text{PO-free}$  fragment (§7.2, §7.7).

4. An honest, technique-organized account of every route tried, the counterexamples that killed the wrong ideas, and the lessons (§7), so that others need not repeat them.
5. A candid methodology report on adversarial AI-assisted mathematics (§9).

The paper is organized as follows. §2 states, in largely non-technical terms, where the problem stands after twenty-five years. §4 gives the history ( $\mathcal{ALCT}_{\text{RCC}}$  from Cohn 1993 through Wessel 2002/2003 and Lutz–Wolter 2006) and carefully distinguishes our object from the neighbouring logics that *are* known undecidable. §5 views the problem as first-order satisfiability and asks whether it falls inside a decidable fragment of first-order logic—it does not, and the deepest reason (totality) previews the central obstruction. §6 explains why the standard undecidability reductions fail. §7—the core of the paper—surveys the approaches *by technique*: naive tableaux and blocking, finite-model and reduction attempts, the split-forest idea, the automata route, the Lean line culminating in F6, and the regular-cover pivot. §8 describes the working reasoners and a GIS example. §9 reports on the AI methodology. §10 concludes with what was learned, what would move the needle, an honest estimate of the chances of closing F6, and the project’s actual wins.

## 2 Current status: where we stand after twenty-five years

*Decidability of  $\mathcal{ALCT}_{\text{RCC5}}$  concept satisfiability remains open* (the closely related  $\mathcal{ALCT}_{\text{RCC8}}$  likewise, but it is not our focus here). A current-literature check (July 2026) found no published resolution; the closest neighbours are decided in the other direction and are discussed in §4. What the present work adds is not an answer but a sharp reduction and a machine-checked surround.

**The positive result—conditional decidability.** *If* the live width of models is bounded by a computable function of the concept (property F6), *then* concept satisfiability is decidable; and the **soundness half of that reduction is machine-checked in Lean 4** (a valid finite certificate provably unfolds to a genuine RCC5 model of the concept, and the finite type abstraction is faithful). What is open is the antecedent F6 itself—equivalently the completeness direction, that every satisfiable concept has a *bounded* finite certificate.

**What is certified** (machine-checked, Lean 4, zero **sorry**):

- the *soundness* pipeline: a valid finite certificate unfolds to a genuine RCC5 model of the concept;
- the *faithfulness* of the finite-type (Hintikka, possibly infinitary) abstraction: satisfiability and abstraction-realizability coincide;
- the *RCC5 normal form, both directions*: every strong-EQ RCC5 network is an ordered-disjoint structure—a strict partial order for PP with a downward-closed disjointness for DR, PO residual—and, conversely, every ordered-disjoint structure is realized by sets under set-RCC5 (GPT-5.6 Pro’s canonical representation, Theorem 15).

**What is strongly supported but not certified:** F6, the open keystone *of the present certificate architecture*—now sharpened to a precise combinatorial dichotomy (does some concept force unbounded “identity-selector minors”?), with the auxiliary uniformization property W2’ shown to *fold into* it. (F6 controls *this* route; its failure would not by itself close off decidability by some other representation—§10.)

### Unconditional by-products (independent of F6):

- the  $\forall\text{PO}$ -free fragment is *decidable*—concepts with no  $\forall\text{PO}$  subformula, still retaining  $\exists\text{PO}$ ,  $\forall\text{DR}$ ,  $\exists\text{DR}$ , and all vertical modalities, so a genuinely expressive spatial fragment (its decision procedure now machine-checked in Lean for the three pure quadrants and the mixed case on a witness; Theorem 10);
- the RCC5 normal form above (both directions certified);
- a working, empirically validated *prototype reasoner* (the cover-tree tableau; no completeness theorem is claimed): cross-validated on 911 concepts against an independent oracle with zero mismatches, reproducing 23 years later the full GIS subsumption taxonomy of Wessel’s original report, and never once contradicted—no counterexample to its verdicts was found anywhere in the campaign;
- the standard undecidability routes shown blocked, with the coincidence obstruction identified as the shared fact.

**The honest bottom line.** After roughly thirty repair rounds and seventeen adversarial reviews the project has reached a genuine *local optimum*: the local algebra is exhausted (and now certified), the soundness side is machine-checked, and the whole remaining difficulty of the *present certificate architecture* is one well-posed lemma. We offer the work not as a resolution but as a two-decade open problem *reduced, honestly and checkably, to its irreducible core*.

**The status levels, at a glance.** Throughout this paper we keep four verification levels distinct; Table 1 is the key, and the prose above and below assigns each result to one level. Conflating them is the single easiest way to misread the project.

## 3 Preliminaries

We fix the object logic precisely; readers familiar with  $\mathcal{ALCI}_{\text{RCC}}$  may skim to Definition 5.

**Syntax.**  $\mathcal{ALCI}_{\text{RCC5}}$  concepts are built from atomic concepts  $A$  and the five RCC5 relations, used as roles, by

$$C ::= A \mid \neg A \mid \top \mid \perp \mid C \sqcap C \mid C \sqcup C \mid \exists R.C \mid \forall R.C, \quad R \in \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}, \text{DR}\}.$$

We work in *negation normal form* (NNF: negation only on atoms) and *absorb inverse roles* through the RCC5 converse— $\exists \text{PP}^-.C = \exists \text{PPI}.C$ ,  $\exists \text{PO}^-.C = \exists \text{PO}.C$ , and so on—so no separate inverse constructor is needed. Let  $\text{cl}(C_0)$  be the Fischer–Ladner closure of the input concept and  $m = |\text{cl}(C_0)|$ .

**Semantics: abstract composition tables, strong EQ.** An interpretation is a pair  $\mathcal{I} = (\Delta, \rho)$  with  $\Delta \neq \emptyset$  and a *total* labelling  $\rho : \Delta \times \Delta \rightarrow \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}, \text{DR}\}$  satisfying: (i) *strong EQ*:  $\rho(x, y) = \text{EQ}$  iff  $x = y$  (so EQ is identity, and between distinct elements only  $\{\text{PP}, \text{PPI}, \text{PO}, \text{DR}\}$  occur); (ii) *converse-closed*:  $\rho(y, x) = \text{conv}(\rho(x, y))$ , where  $\text{PP}^{-1} = \text{PPI}$ ,  $\text{PO}^{-1} = \text{PO}$ ,  $\text{DR}^{-1} = \text{DR}$ ; and (iii) *composition-closed*:  $\rho(x, z) \in \text{comp}(\rho(x, y), \rho(y, z))$  for all  $x, y, z$ , over the RCC5 composition table (Table 3). Concept extensions are standard, reading a role  $R$  as the pair set  $\{(x, y) : \rho(x, y) = R\}$ ; the concept  $C_0$  is *satisfiable* if  $x \in C_0^{\mathcal{I}}$  for some  $\mathcal{I}$  and some  $x \in \Delta$ . Thus

| Status                                                        | Examples                                                                                                                                                                                                                                                                                                                                                                                                                  | What the label means                                       |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <i>Lean-certified</i>                                         | certificate soundness; finite- <i>type</i> Hintikka faithfulness; the full (both-direction) ordered-disjoint normal form; the $\forall$ PO-free lift core <i>and its decision procedure</i> for the three pure quadrants ( $\exists$ DR/PO/EQ, $\exists$ PP, $\exists$ PPI) plus the mixed-case decision on a witness                                                                                                     | kernel-checked theorem names with a stated axiom footprint |
| <i>Theorem-level</i> (not end-to-end certified)               | the $\forall$ PO-free fragment’s specific $K(C_0)$ closed-form bound; the identity-selector characterization; the $W2'$ refinement; the local extension / cross-policy results                                                                                                                                                                                                                                            | a proof in a named work package, often with finite probes  |
| <i>Lean-certified</i> (2026-08-29), cold-reviewed three times | a decision procedure for the <i>whole</i> $\forall$ PO-free fragment, mixed quadrant included, on <i>raw</i> (not pre-normalized) input and under <i>concrete set</i> semantics ( <code>decidableFSat</code> , <code>decidableSat_cone</code> , <code>decidableSetSat</code> ), with fragment membership its only hypothesis; and a one-sided UNSAT test for the <i>full</i> logic ( <code>coneScheme_unsat_full</code> ) | see §7.3                                                   |
| <i>Empirical</i>                                              | the cover-tree tableau; the cross-validation suites; the finite model verifier                                                                                                                                                                                                                                                                                                                                            | test evidence only; not a decision theorem                 |
| <i>Open</i>                                                   | static F6; completeness of the chosen certificate scheme; the automatic-cover property; full decidability                                                                                                                                                                                                                                                                                                                 | no proof claimed                                           |

Table 1: The four verification levels used throughout. The paragraphs of §2 assign each result to a level; this table is the at-a-glance key.

a model is a complete graph carrying a composition-closed RCC5 edge-labelling; *models may be infinite*. This is precisely the *abstract* (composition-table) semantics—not a semantics of actual regions in a topological space. We fix the *strong* EQ semantics throughout (the standard convention for this problem; strong versus weak EQ is discussed below). The decision problem is: given  $C_0$ , is it satisfiable?

**The sister logic  $\mathcal{ALCI}_{\text{RCC8}}$ .**  $\mathcal{ALCI}_{\text{RCC8}}$  is defined identically, over the eight RCC8 relations  $\{\text{DC}, \text{EC}, \text{PO}, \text{EQ}, \text{TPP}, \text{NTPP}, \text{TPPI}, \text{NTPPI}\}$ , which *refine* RCC5 by splitting DR into DC (disconnected) and EC (externally connected, i.e. touching at the boundary), and PP, PPI into tangential and non-tangential parts (TPP, NTPP and TPPI, NTPPI); PO and EQ are unchanged. Its abstract composition-table semantics is the exact analogue over the RCC8 composition table, and RCC8 likewise has the patchwork property.  $\mathcal{ALCI}_{\text{RCC5}}$  is our focus; we invoke  $\mathcal{ALCI}_{\text{RCC8}}$  only where it matters—its EXPTIME-hardness (§4) and the domino attempts (§6), which turn on the extra TPP/NTPP distinction.

**The RCC granularity hierarchy.**  $\mathcal{ALCI}_{\text{RCC5}}$  and  $\mathcal{ALCI}_{\text{RCC8}}$  sit atop a family of five logics  $\mathcal{ALCI}_{\text{RCC}k}$ ,  $k \in \{1, 2, 3, 5, 8\}$ , whose relation sets *coarsen* as  $k$  decreases [9]:  $\mathcal{ALCI}_{\text{RCC}1}$  has the single relation SR (“spatially related”);  $\mathcal{ALCI}_{\text{RCC}2}$  has  $\{\text{DR}, \text{O}\}$  (with O = “overlapping”); and



$\mathcal{ALCI}_{\text{RCC3}}$  has  $\{\text{DR}, \text{ONE}, \text{EQ}\}$  (with  $\text{ONE}$  = “overlapping but not equal”). Each coarser set is definable in the finer one— $\text{ONE} = \{\text{PP}, \text{PPI}, \text{PO}\}$ ,  $\text{O} = \{\text{ONE}, \text{EQ}\}$ ,  $\text{SR} = \{\text{DR}, \text{O}\}$ —so the family forms a granularity lattice. The three coarsest logics are *decidable* (Wessel [9]);  $\mathcal{ALCI}_{\text{RCC5}}$  and  $\mathcal{ALCI}_{\text{RCC8}}$  are the open cases this paper concerns. The full landscape is Table 2 (§4).

**Strong versus weak EQ.** Two semantics for EQ appear in the literature [9]: the *strong* one,  $\text{EQ}^{\mathcal{I}} = \text{Id}(\Delta)$  (EQ is exactly identity), and the *weak* one,  $\text{Id}(\Delta) \subseteq \text{EQ}^{\mathcal{I}}$  (EQ contains identity but may relate distinct, mutually indistinguishable elements). We fix the *strong* semantics throughout—the standard choice for this problem, and the more expressive one: Wessel observed that under strong EQ, from  $\mathcal{ALCI}_{\text{RCC3}}$  upward the *difference role*  $R_D =_{\text{def}} N_R \setminus \{\text{EQ}\}$  becomes definable, so *nominals* are encodable ( $i \sqcap \forall R_D. \neg i$  forces  $i$  to name a single element)—a reminder of how narrow the margin to undecidability is (adding a  $\downarrow$ -binder would cross it, cf. §4.3). The choice costs no generality for decidability: Wessel showed decidability under the weak semantics would carry over to the strong one, and conversely a weak-EQ model collapses to a strong one by internalizing EQ as identity in the TBox, so the two satisfiability problems are inter-reducible.

**The two axes, made precise.** Call PP, PPI *vertical* (nesting) and DR, PO *horizontal* (side-by-side). The composition table makes the axes behave oppositely, and this asymmetry is the root of the whole difficulty.

**Proposition 1** (Determination is vertical). *Of the sixteen compositions of proper (non-EQ) relations, exactly four are determining (return a singleton):*

$$\text{comp}(\text{DR}, \text{PPI}) = \{\text{DR}\}, \text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}, \text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}, \text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}.$$

*Every one has a vertical (PP or PPI) argument, and PO is never the value of any determining composition. (Exhaustive over Table 3; machine-verified.)*

So no *single* composition step forces a horizontal value—the four determining steps are all vertical. This must not be overread:

**Remark 2** (Single-step versus joint forcing). Proposition 1 concerns *one* length-two path; it does *not* say horizontal edges are always free. The *intersection* of several composition constraints can pin a horizontal edge to a single value:  $\text{comp}(\text{DR}, \text{PO}) = \{\text{DR}, \text{PO}, \text{PP}\}$  and  $\text{comp}(\text{PPI}, \text{PO}) = \{\text{PPI}, \text{PO}\}$  intersect in  $\{\text{PO}\}$ , so an edge forced through two different intermediates can be *compelled* to be PO (machine-checked, wp85); indeed every proper relation—DR, PO, PP, PPI—is reachable as such a joint singleton, so PO is not special among them. Consequently “shadow” and “live” (Definition 5) must be read against the *joint* closure of all constraints, never single-path forcing. What *composition* genuinely never forces is a merge: no intersection of composition cells whose two premise labels are *proper* (non-EQ) relations ever narrows to  $\{\text{EQ}\}$  (wp85), so composition can only force occurrences *apart* (a shared horizontal anchor excludes EQ), never *together*. Concept *labels*, by contrast, *can* force a *local* merge: the satisfiable one-level gadget of §6.2 forces its two coloured witnesses to coincide and remains satisfiable. What fails is *propagating* that isolation recursively—the discipline that would tile a grid collapses the construction to unsatisfiability. So the coincidence obstruction is really a *dichotomy* at the level of the would-be grid: a grid merge is either *unforced* (the branches never join) or *fatal* (recursion collapses it to UNSAT), never a clean forced grid—not a blanket “EQ cannot be forced,” and not a claim that no merge is ever satisfiable. That one-sidedness—composition forces apart but not (usefully) together—is the real horizontal asymmetry.



There is one further asymmetry, and it is the one the split-forest presentation (§7.4) turns to its advantage: disjointness is inherited *downward* but not upward.

**Proposition 3** (Half-determinism of disjointness). *If  $x \text{ DR } y$  then every part of  $x$  is disjoint from every part of  $y$ :  $\text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$  forces  $x' \text{ DR } y$  for every  $x' \text{ PP } x$ , and  $\text{comp}(\text{DR}, \text{PPI}) = \{\text{DR}\}$  pushes this down the  $y$ -side as well. Upward the inference fails: a superpart of a disjoint region may be disjoint from, overlap, or contain the other region— $\text{comp}(\text{PPI}, \text{DR}) = \{\text{PPI}, \text{PO}, \text{DR}\}$  is not a singleton, so nothing is forced. Disjointness thus propagates deterministically to descendants in the PP-order, and to nothing else—the downward-closed disjointness that the RCC5 normal form (Theorem 15, certified) makes exact.*

This “half-determinism” (deterministic downward, free upward) is what lets a finite presentation dispatch arbitrarily wide DR crowds for free (§7.4): a single high DR edge shadows every disjointness beneath it, leaving only PO and the *upward* direction as genuine branching. Vertical recurrence, moreover, always folds:

**Lemma 4** (No horizontal recurrence). *Form the composition-path subset automaton whose states are the reachable relation sets, with  $\text{Rel}(\varepsilon) = \{\text{EQ}\}$  and  $\text{Rel}(w a) = \bigcup_{r \in \text{Rel}(w)} \text{comp}(r, a)$ . Restrict to states not containing EQ, and to non-EQ labels (an EQ label is a self-loop on every state, since EQ is the composition identity). Then no cycle uses a horizontal (DR/PO) label: every cycle is built from vertical (PP/PPI) steps. Hence any support recurrence that never folds by an equality is eventually vertical, and is absorbed by vertical looping rather than by a horizontal crowd. (Exhaustive finite computation; machine-verified, wp38.)*

Proposition 1 and Lemma 4 explain why infinite *depth* is never the obstacle: forced structure is vertical, and vertical chains loop. Everything hard is horizontal—and, crucially, while *individual* horizontal edges can be pinned (by the joint closure of several constraints, Remark 2), no concept is known to *force an unboundedly wide rigid* horizontal pattern. Whether one can is exactly F6. A decision procedure must record only the horizontal branching it cannot reconstruct:

**Definition 5** (Shadows, live width, and F6). In a model, or in a finite presentation of one, an edge is a *shadow* if its RCC5 value is forced by the *joint* composition-closure of the edges already present—which may be a single triangle ( $x \text{ PP } y, y \text{ DR } z$  force  $x \text{ DR } z$  by  $\text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$ ) or the intersection of several (Remark 2); a shadow costs a decision procedure nothing. The *live width* at an interface is the number of non-shadow edges there—the genuine branching that must be recorded. Property **F6** states: for every satisfiable  $C_0$ , the live width of some presentation of a model of  $C_0$  is bounded by a computable function of  $C_0$ .

## 4 History of $\mathcal{ALCI}_{\text{RCC}}$ and why the known results do not settle it

### 4.1 Cohn’s 1993 proposal

The idea of combining description logics with qualitative spatial reasoning via modal accessibility relations was first proposed by Cohn [1], who considered treating the RCC8 relations (or subsets thereof) as modalities in a multi-modal logic. The framework allowed spatial constraints to be expressed as modal formulas—e.g.,  $\langle \text{PP} \rangle C$  for “some proper part satisfies  $C$ ”—but Cohn left all decidability and complexity questions open.

| Logic                                                 | Base relations $N_R$            | Concept satisfiability                    |
|-------------------------------------------------------|---------------------------------|-------------------------------------------|
| $\mathcal{ALCI}$ (no RCC)                             | arbitrary roles                 | PSPACE-complete (bare); EXPTIME w/ TBoxes |
| $\mathcal{ALCI}_{\text{RCC1}}$                        | {SR}                            | decidable; NP-complete ( $\equiv$ S5)     |
| $\mathcal{ALCI}_{\text{RCC2}}$                        | {DR, O}                         | decidable; NP-hard, in PSPACE             |
| $\mathcal{ALCI}_{\text{RCC3}}$                        | {DR, ONE, EQ}                   | decidable; NP-hard                        |
| $\mathcal{ALCI}_{\text{RCC5}}, \forall\text{PO-free}$ | RCC5 without $\forall\text{PO}$ | <b>decidable</b> (this paper, Thm 10)     |
| $\mathcal{ALCI}_{\text{RCC5}}$ (full)                 | {DR, PO, EQ, PP, PPI}           | <b>open</b> ; PSPACE-hard                 |
| $\mathcal{ALCI}_{\text{RCC8}}$                        | RCC8 (eight relations)          | <b>open</b> ; EXPTIME-hard                |

Table 2: The complexity landscape of the  $\mathcal{ALCI}_{\text{RCC}}$  family (concept satisfiability, strong EQ).  $\text{RCC}k$  relation sets coarsen as  $k$  decreases;  $\mathcal{ALCI}_{\text{RCC1}}\text{--}\mathcal{ALCI}_{\text{RCC3}}$  were shown decidable by Wessel [9], while  $\mathcal{ALCI}_{\text{RCC5}}$  and  $\mathcal{ALCI}_{\text{RCC8}}$  are the standing open cases.

## 4.2 Wessel’s $\mathcal{ALCI}_{\text{RCC}}$ family (2002/2003)

Wessel [7, 9, 5] gave the first rigorous treatment of the combination, introducing the  $\mathcal{ALCI}_{\text{RCC}}$  family: description logics  $\mathcal{ALCI}_{\text{RCC}k}$  ( $k = 1, 2, 3, 5, 8$ ) extending  $\mathcal{ALCI}$  with role boxes derived from the RCC composition tables. In an  $\mathcal{ALCI}_{\text{RCC}k}$  interpretation, every pair of domain elements is assigned exactly one  $\text{RCC}k$  base relation, and the composition table is enforced globally (*strong EQ semantics*: EQ is identity). Models are complete graphs with  $\text{RCC}k$  edge-colorings.

Wessel established the decidability and complexity landscape summarized in Table 2. The three coarsest logics are *decidable*:  $\mathcal{ALCI}_{\text{RCC1}}$  is essentially the modal logic S5 (hence NP-complete);  $\mathcal{ALCI}_{\text{RCC2}}$  and  $\mathcal{ALCI}_{\text{RCC3}}$  are decidable because their composition tables are deterministic or near-deterministic, admitting reduction to standard DL techniques (for  $\mathcal{ALCI}_{\text{RCC3}}$  under strong EQ, via the definable difference role). The two finest logics— $\mathcal{ALCI}_{\text{RCC5}}$  (PSPACE-hard) and  $\mathcal{ALCI}_{\text{RCC8}}$  (EXPTIME-hard)—were left as the central open problem.

In a series of companion reports, Wessel also mapped out the decidability landscape for description logics with general composition-based role inclusion axioms:

- $\mathcal{ALC}_{\mathcal{RA}}$  (arbitrary role axioms) is **undecidable**, via reduction from the Post Correspondence Problem [6].
- $\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$  (non-disjoint-roles variant) is also **undecidable**, via reduction from non-empty CFG intersection [3].
- $\mathcal{ALC}_{\mathcal{RASG}}$  (admissible role boxes: deterministic, functional, complete, and associative) is **decidable** (EXPTIME-complete, via reduction to  $\mathcal{ALC}$  with general TBoxes) [4]; adding unqualified number restrictions ( $\mathcal{ALCN}_{\mathcal{RASG}}$ ) makes it undecidable again.

$\mathcal{ALCI}_{\text{RCC5}}$  sits precisely in the gap: its role box is non-deterministic (unlike  $\mathcal{ALC}_{\mathcal{RASG}}$ ) but has the patchwork property (unlike  $\mathcal{ALC}_{\mathcal{RA}}/\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$ ).

**Remark 6** (Key difficulties).  $\mathcal{ALCI}_{\text{RCC5}}$  has none of the properties that standard DL decidability proofs rely on: (i) no tree model property (models are complete graphs), (ii) no finite model property (some satisfiable concepts require infinite models), (iii) a non-deterministic role box (multi-valued composition entries).

For reference, the full RCC5 composition table is given in Table 3. Reading: row =  $S(b, c)$ , column =  $R(a, b)$ , entry = possible relations for  $(a, c)$ .

| comp          | DR( $a, b$ ) | PO( $a, b$ ) | EQ( $a, b$ ) | PPI( $a, b$ )   | PP( $a, b$ ) |
|---------------|--------------|--------------|--------------|-----------------|--------------|
| DR( $b, c$ )  | *            | DR, PO, PPI  | DR           | DR, PO, PPI     | DR           |
| PO( $b, c$ )  | DR, PO, PP   | *            | PO           | PO, PPI         | DR, PO, PP   |
| EQ( $b, c$ )  | DR           | PO           | EQ           | PPI             | PP           |
| PP( $b, c$ )  | DR, PO, PP   | PO, PP       | PP           | PO, EQ, PP, PPI | PP           |
| PPI( $b, c$ ) | DR           | DR, PO, PPI  | PPI          | PPI             | *            |

Table 3: The RCC5 composition table (\* = all five relations).

Key composition facts exploited in the cover-tree theory (§7.4, §8):  $\text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$  and  $\text{comp}(\text{DR}, \text{PPI}) = \{\text{DR}\}$  (DR rigidity);  $\text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$  and  $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$  (transitivity); and  $\text{comp}(\text{DR}, \text{PO}) = \{\text{DR}, \text{PO}, \text{PP}\}$  (PP can be *forced* by composition and universal filtering; see the forced-composition examples in §7.1).

### 4.3 Wessel’s grid constructions and the coincidence obstruction

In his investigation of  $\mathcal{ALCI}_{\text{RCC8}}$  [9], Wessel made two important discoveries:

**1. Even-odd chains (Section 4.4.2).** The TPP/NTPP distinction in RCC8 allows the construction of *functional successor chains*: each node alternates between *even* and *odd*, and a node can distinguish its direct TPPI-successor from all indirect ones by parity. Wessel used this to encode a binary  $n$ -bit counter, proving  $\mathcal{ALCI}_{\text{RCC8}}$  is EXPTIME-hard.

**2. The grid coincidence obstruction (Section 4.4.3, Figure 11).** Wessel constructed explicit RCC8 networks isomorphic to  $n \times n$  grids, using TPP for horizontal and vertical successors. However, he identified a fundamental obstacle to encoding domino tilings: the *coincidence condition*  $R_X \circ R_Y = R_Y \circ R_X$  (east-of-north equals north-of-east) cannot be enforced by concept terms. The composition table allows  $\{\text{PO}, \text{EQ}, \text{TPP}, \text{NTPP}, \text{TPPI}, \text{NTPPI}\}$  at the coincidence point—too many options. Wessel concluded:

“Even though these infinite models do exist, we have found no way to *enforce* the depicted model(s) by means of a concept term.”

Wessel further observed that adding a hybrid logic binding operator  $\downarrow$  (nominals) would immediately yield undecidability, confirming that the coincidence condition is the *precise boundary*. His explicit grid frame is reproduced here as Figure 1, and the merge obstruction that blocks enforcing it as Figure 2.

Wessel thus independently developed, three to four years before Lutz–Wolter, the RCC8 description-logic analogues of these ingredients—the even-odd chain, an explicit two-dimensional TPP/NTPP grid, and the first identification of the coincidence gap as the decisive obstacle. Lutz–Wolter later obtained undecidability by a *different* route—a linearized tiling enumeration (their §5, following Marx–Reynolds) that sidesteps the two-dimensional coincidence rather than enforcing it—so this is shared lineage and prior identification, not a dependence of their proof on Wessel’s constructions.

### 4.4 Lutz & Wolter (2006)

Lutz and Wolter [10] studied *modal logics of topological relations*, denoted  $L_{\text{RCC8}}$ ,  $L_{\text{RCC5}}$ , etc. These have the same *syntax* as  $\mathcal{ALCI}_{\text{RCC}}$  but are interpreted over *topological models*: the domain

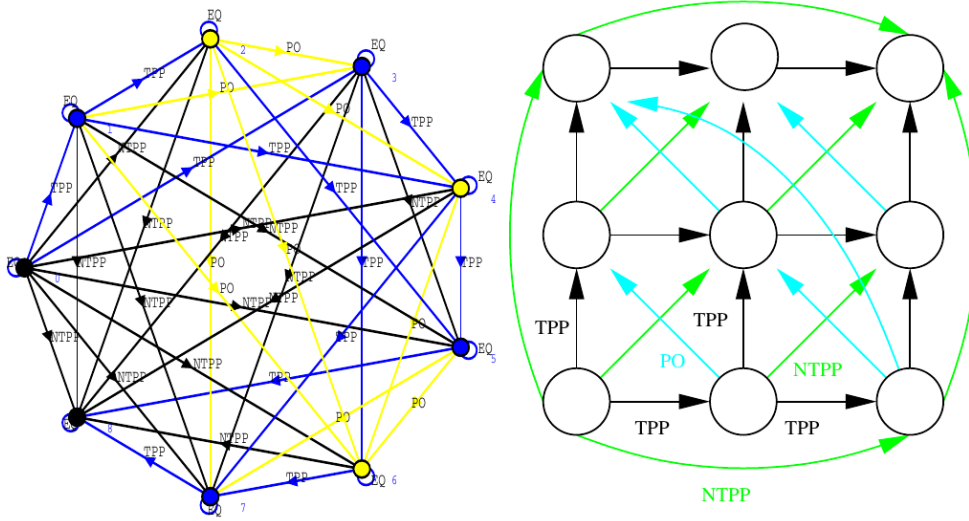


Figure 1: Wessel’s original  $\mathcal{ALCI}_{RCC8}$  grid construction, reproduced *directly from report7.pdf* (2002/2003), Figure 10 (Section 4.4.3): a complete-graph RCC8 “frame” (left) that is isomorphic to a  $3 \times 3$  grid (right) and extends to  $n \times n$ . Horizontal and vertical successors are TPP edges (the intended roles  $R_X, R_Y$ ); PO (cyan) and NTPP (green) are the induced diagonal and long-range relations. Such grid models *exist*—but, as the same report already argued, no concept term can *enforce* them. (The frame was *found by a Lisp enumerator* and rendered with CLIM under MCL, not drawn by hand.) The construction and the obstruction are Wessel’s, two decades before this AI-assisted work.

elements are regular closed subsets of a topological space (typically  $\mathbb{R}^n$ ). They proved:

- $L_{RCC8}$  is **undecidable** (their §5: a linearized NTPP-chain enumeration of tile positions, following Marx–Reynolds, whose faithfulness relies on the geometry of  $\mathbb{R}^n$ ; no two-dimensional grid coincidence is enforced).
- $L_{RCC5}(RS^\exists)$  is **undecidable** (reduction from  $S5^3$  via supremum regions).
- $L_{RCC5}(RS)$  is **open**. Lutz and Wolter explicitly wrote (p. 31): “Perhaps the most interesting candidate is  $L_{RCC5}(RS)$  [...] to which the reduction exhibited in Section 8 does not apply.”

The logic  $\mathcal{ALCI}_{RCC5}$  is exactly  $L_{RCC5}(RS)$ : arbitrary complete-graph models with composition-table semantics. The problem has been recognized as open by leading researchers since 2006.

**The route they suggested, and how it differs from the one taken here.** Their concluding section asks for “a decidable and still useful variant” of the logics they study, and canvasses where one might be found. Besides  $L_{RCC5}(RS)$  itself, they write that “other candidates could be obtained by modifying the set of relations, e.g. giving up some of them,” and add that “it has, for example, been argued that dropping PO still results in a useful formalism for applications in geographic information systems” [10, p. 31].

It is worth being exact about what that is and is not, because it is the closest thing in the literature to the fragment established here. The sentence about dropping PO discharges the *usefulness* half of their two-part desideratum, not the decidability half: it is a claim about applications, it carries no citation, and it is accompanied by no proof, no proof sketch, and no conjecture that a PO-less

language would be decidable. The technical progress they point to in the same breath, Shapirovsky and Shehtman [8], concerns modal logics with a *single* modality for an inclusion relation over region structures, doubling as relativistic temporal logics — a substantially smaller language than RCC5 without PO, which still carries DR, PP and PPI. So far as we can determine, no decidability argument for any PO-restricted RCC5 language appears there or since.<sup>1</sup>

The fragment established here differs from their suggestion in two ways, and both matter. First, it is a strictly *weaker* restriction: PO is not removed from the language. Existential quantification  $\exists \text{PO}.C$  — “overlaps something satisfying  $C$ ” — is retained in full, and only the universal  $\forall \text{PO}.C$  is excluded, so the fragment is properly larger than the one their remark contemplates. Second, and structurally more important, the fragment does *not* obtain decidability by recovering a finite model property. The concept  $\exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$  is  $\forall \text{PO}$ -free and satisfiable with no finite model (`cinf_pofree`, `cinf_satisfiable`), so every model of it is infinite; decidability comes instead from a finite *presentation* of possibly-infinite models (§7.3). And the reason the restriction is exactly the right one is the residual-PO default: a composition cell containing PO can always be discharged by putting the pair at PO unless some formula can object, and the only formula that can is  $\forall \text{PO}.D$ . Dropping PO wholesale would buy that property too, but at more cost than is necessary.

**Prior consideration of restricted fragments, by the present author.** Report7 [9] contains a section headed “Restricted Decidable Fragments of  $\mathcal{ALCI}_{\text{RCC5}}$  and  $\mathcal{ALCI}_{\text{RCC8}}$ ?” (§5.3), which is prior art for the general idea and should be recorded as such. It canvasses two restrictions: one derived from  $\mathcal{ALCRP}(D)$ , rejecting every concept whose negation normal form contains a quantifier alternation  $\exists R.\dots\forall S.\dots$  or  $\forall R.\dots\exists S.\dots$ ; and one admitting only Boolean concepts inside universal value restrictions. Both are strictly stronger than  $\forall \text{PO}$ -freeness — they constrain *every* role, and forbid the nesting the present fragment permits — and both work, as that section says, by restoring the finite model property. Neither is the  $\forall \text{PO}$ -free fragment, and by the witness above neither technique could yield it.

**On priority and citation.** Two honest caveats, in both directions. First, Wessel proved *neither* decidability nor undecidability of  $\mathcal{ALCI}_{\text{RCC5}}$  or  $\mathcal{ALCI}_{\text{RCC8}}$ : report7 constructs the grid models and pins the coincidence obstruction as the precise boundary, but stops there, explicitly leaving both problems open. The contribution was to *locate* the wall, not to breach it. Second, and less comfortably: the Lutz–Wolter paper does not cite report7, although it was a publicly available official technical report of the University of Hamburg (2002/2003) that already developed the  $\mathcal{ALCI}_{\text{RCC}}$  description logics, the even-odd counter construction, the explicit two-dimensional TPP/NTTP grid, and the coincidence obstruction as the decisive obstacle. We record this as a matter of *priority*, not of dependence—their undecidability proof takes a genuinely different (topological, linearized) route and in no way relies on report7—but the overlap in ingredients was substantial and the material was findable.

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<sup>1</sup>One might be tempted to read support into the fact that PO occurs exactly once in their RCC5 undecidability reduction (§8) — as the box [PO] in their formula (8.1), forcing the representing regions to be pairwise disconnected — and never in existential position, so that removing  $\forall \text{PO}$  alone suffices to break it. That reading would be a mistake, and we record the observation only because it makes their choice of PO look informed rather than arbitrary. Breaking one reduction is no evidence of decidability: undecidability proofs are not unique, and nothing excludes a different argument that never mentions PO. The entailment in fact runs the other way and in the uninteresting direction — the fragment is decidable, hence *no* undecidability reduction succeeds on it, theirs included. Their reduction moreover fails for  $L_{\text{RCC5}}(RS)$  on independent grounds, requiring closure under suprema ( $RS^\exists$ ), which is why they list that logic as the open candidate in the first place.

| Approach                                                        | $\forall PP.C?$ | $\exists DR.D?$ | Quantify over relations? |
|-----------------------------------------------------------------|-----------------|-----------------|--------------------------|
| Concrete domains, e.g. $\mathcal{ALC}(\text{RCC8})$             | no              | no              | no                       |
| Composition-based role boxes ( $\mathcal{ALCI}_{\text{RCC5}}$ ) | <b>yes</b>      | <b>yes</b>      | <b>yes</b>               |

Table 4: Concrete domains versus composition-based role boxes. Only the latter can *quantify over* spatial relations, which is exactly what puts  $\mathcal{ALCI}_{\text{RCC5}}$  outside the reach of the concrete-domain results.

#### 4.5 Concrete-domain approaches, and why they do not settle it

A natural objection is that decidability of description logics with RCC constraints is surely already known. It is—but for a *different*, expressively incomparable formalism: the *concrete-domain* approach, in which spatial relations constrain *functional attributes* of individuals rather than serving as roles. The distinction is decisive (Table 4). A concrete domain reasons about the spatial *attributes* of elements;  $\mathcal{ALCI}_{\text{RCC}}$  instead makes the spatial relations *themselves* the roles, and so permits *quantification over* them—concepts such as  $\forall PP.C$  (“every proper part satisfies  $C$ ”) and  $\exists DR.D$  (“some discrete region satisfies  $D$ ”)—which the concrete-domain setting cannot express.

The known decidability results all sit on the concrete-domain side. Lutz and Milićić [11] showed  $\mathcal{ALC}$  with  $\omega$ -admissible concrete domains (which include RCC5 and RCC8) decidable, without inverse roles; Baader and Rydval [12] revisited and generalized the  $\omega$ -admissibility conditions (and strengthened the matching undecidability results); Borgwardt, De Bortoli and Koopmann [13] pinned  $\mathcal{ALC}(\mathcal{D})$  ontology consistency at EXPTIME-complete for  $\omega$ -admissible  $\mathcal{D}$  ( $\mathcal{ALC}$  only, no inverse); and Demri and Gu [14] recently extended the constraint-automaton method to *inverse* roles and functional role names with EXPTIME membership—the closest published result to  $\mathcal{ALCI}_{\text{RCC}}$ , yet still within the concrete-domain formalism. None settles  $\mathcal{ALCI}_{\text{RCC5}}$  or  $\mathcal{ALCI}_{\text{RCC8}}$ : the composition-based role box allows quantification over the spatial relations themselves, and the complete-graph semantics is not a concrete-domain constraint. The two lines are complementary—attributes versus relationships—which is why twenty years of progress on the former left the latter open.

### 5 Decidable fragments of first-order logic, and why none captures it

$\mathcal{ALCI}_{\text{RCC5}}$  satisfiability is, quite literally, a *first-order* satisfiability problem. The five relations are binary predicates; the composition-table semantics is a finite set of first-order axioms  $T_{\text{RCC5}}$ —exactly one atom on each ordered pair of distinct elements, EQ as identity, converse coherence, and, for every pair of atoms  $r, s$  and every  $t \notin \text{comp}(r, s)$ , the *forbidden triangle*  $\forall x y z \neg(R_r(x, y) \wedge R_s(y, z) \wedge R_t(x, z))$ —while the concept  $C_0$  becomes a sentence  $\exists x \text{ST}(C_0, x)$  under the standard modal translation. Thus  $C_0$  is satisfiable iff  $T_{\text{RCC5}} \cup \{\exists x \text{ST}(C_0, x)\}$  has a model. If this theory fell inside a *decidable fragment* of first-order logic we would be done—a natural first thing to try, and, at coarser granularity, exactly how the decidable family members fell: report7 settles  $\mathcal{ALCI}_{\text{RCC3}}$  by precisely this route (below). At RCC5 it does not work; the surface reasons vary by fragment, but the deepest one—the one that defeats even the fragments tailor-made for relational theories like this—is the same totality that underlies F6.

Two invariants of  $\mathcal{ALCI}_{\text{RCC5}}$  models constrain which fragment could contain it.

- **No finite model property.** An ascending PP-tower ( $\exists \text{PPI}.\top \sqcap \forall \text{PPI}.\exists \text{PPI}.\top$ , with PP a strict



order) is satisfiable only in infinite models. Any fragment with the finite model property is excluded outright.

- **Complete Gaifman graph.** Totality forces every two distinct elements to be related, so the Gaifman graph<sup>2</sup> of every model is *complete*: a PP-chain of length  $n$  already has treewidth  $n-1$  and clique number  $n$  (wp78). Any fragment whose decidability rests on models of bounded treewidth—the whole guarded family—is excluded as well.

Against these, the standard fragments fall as follows.

*Two variables* ( $\text{FO}^2$ ). This is how the coarser family members actually fell: report7 proves  $\mathcal{ALCI}_{\text{RCC3}}$  (strong EQ) decidable by translating concepts, via the standard translation, into *two-variable* first-order logic with equality [9]—possible there because the RCC3 role theory needs no ternary composition axiom at all (report7’s role theory is exactly-one plus converse symmetry, all two-variable; the RCC3 table constrains no triple of distinct elements). At RCC5 the route is excluded twice over: the composition axiom is now genuinely ternary ( $\text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$  constrains distinct triples), and  $\text{FO}^2$  has the finite model property [15], which  $\mathcal{ALCI}_{\text{RCC5}}$  lacks. Worse, the ternary step is into hostile territory—the three-variable fragment  $\text{FO}^3$  is already *undecidable* (it captures the equational theory of relation algebras [16])—so variable-counting alone offers no route: any decidability must come from the *shape* of the axioms, not their arity.

*Prefix classes* (Bernays–Schönfinkel–Ramsey and relatives). The standard translation of nested modalities produces unbounded  $\forall\exists$  alternation, so no bounded quantifier-prefix class applies; these classes also have the finite model property.

*The guarded fragment and its relatives.* This is the interesting near-miss, and it is worth being precise about where it breaks. The concept part is *guarded* outright—the modal translation is the paradigm guarded formula, each quantifier relativized by its role atom ( $\exists y(R_{\text{PP}}(x, y) \wedge \dots)$ ) [17]. The composition constraint, in forbidden-triangle form, is *clique-guarded*: in  $R_r(x, y) \wedge R_s(y, z) \wedge R_t(x, z)$  every pair of variables co-occurs in an atom—exactly a clique guard—so composition-closure lives in Grädel’s clique-guarded fragment [18]; converse and mutual-exclusion are guarded too. In other words *almost* the entire theory is clique-guarded. What is not—and cannot be—is **totality**: “every two elements are related” is the archetypal non-guarded assertion (it is not invariant under the guarded bisimulation these fragments are built around). And totality is not a technicality one may drop: it is what forces the complete Gaifman graphs above, which destroy the generalized tree-model property on which the guarded family’s decidability rests. The guarded route thus fails for exactly the reason wp78 records—and note it is *totality*, not the ternary composition axiom, that does the damage; the composition axiom is the tamer half.

**Remark 7** (On the two renderings of the composition table). The *natural* first-order rendering of a table cell is not the forbidden triangle but the implication

$$\forall x y z \left( R_r(x, y) \wedge R_s(y, z) \rightarrow R_{t_1}(x, z) \vee \dots \vee R_{t_n}(x, z) \right), \quad \{t_1, \dots, t_n\} = \text{comp}(r, s),$$

and it is instructive that this form is *less* guarded, not more. Its guard  $R_r(x, y) \wedge R_s(y, z)$  is a *path*, not a clique: the pair  $\{x, z\}$  the consequent speaks about never co-occurs in a guard atom—indeed for  $r = s = t_1 = \text{PP}$  the sentence *is* transitivity,  $\forall xyz (R_{\text{PP}}(x, y) \wedge R_{\text{PP}}(y, z) \rightarrow R_{\text{PP}}(x, z))$ , the textbook non-guardable shape (and the one behind the transitivity undecidability cited below).

<sup>2</sup>The *Gaifman graph* of a relational structure is the undirected graph on its elements in which two distinct elements are joined whenever they occur together in some tuple of some relation—for binary relations, whenever  $R(x, y)$  or  $R(y, x)$  holds for some  $R$ . It is the standard way to speak of the “locality” or “sparsity” of a model: bounded degree, bounded treewidth, and similar tameness conditions are all read off it. Here, because every pair of elements carries an RCC5 atom, this graph is *complete*—maximally non-local.



The contrapositive forbidden triangles are clique-guarded precisely because the offending third edge moves *into the guard*. The two renderings are equivalent only modulo exactly-one: the implication additionally *forces* an atom on  $(x, z)$ —a relativized piece of totality—and the negative rewriting is available at all only because the relation alphabet is finite and exhaustive (for a single transitive relation over an open vocabulary no such rewriting exists). Nor is the rewriting a loophole: within the clique-guarded fragment the triangle sentences are *strictly weaker* than transitivity, since a model may leave  $(x, z)$  atom-free; the equivalence broker is totality, which is exactly the axiom the fragment cannot have. Nothing downstream is sensitive to the choice—both are finite first-order renderings, interchangeable over  $T_{\text{RCC5}}$ , so the  $\Pi_1^0$  placement (§7.6) is unaffected—but the diagnosis is: written with implications, composition *looks* unguarded for a repairable, surface reason; written with triangles, every axiom of  $T_{\text{RCC5}}$  dresses in guards *except* totality, which is the honest accounting.

*Other fragments.* The fluted fragment requires predicate arguments to form a suffix of the quantifier sequence; the composition atom  $R_t(x, z)$  skips  $y$ , so it is not fluted. The unary-negation and guarded-negation fragments are a subtler near-miss: negation of *sentences* is unrestricted there, so the forbidden triangles themselves *are* expressible (each is the negation of an existential-positive sentence)—but both fragments have the finite model property [19], which  $\mathcal{ALCI}_{\text{RCC5}}$  lacks, and totality is again out of reach (its inner negations carry two free variables and no atom to guard them). And the one extension that would embrace the vertical structure directly—the guarded fragment *with transitive relations*, since PP, PPI are transitive ( $\text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$ )—is *undecidable* once transitivity is used outside guards, as it must be here [20], so even the most sympathetic extension supplies no decision procedure.

**The lesson.** The surface reasons vary—arity, prefix shape, atom shape, the finite model property—but those are boundary mismatches that fragment engineering could in principle negotiate. The failure that matters is the guarded family’s, and there the culprit is *totality*, i.e. the complete Gaifman graph: the one axiom of  $T_{\text{RCC5}}$  no guard can carry, and the one obstruction no choice of fragment removes, because it is a property of the *models*, not of the axioms’ syntax. It is precisely the obstruction F6 names from the width side and **wp78** from the cover side; the FO-fragment view rediscovers it from classical model theory. It also tells us which tools are the wrong ones: guardedness and locality (Gaifman, Hanf) are exactly what totality defeats, so the productive machinery is the totality-tolerant kind—Ehrenfeucht–Fraïssé / back-and-forth and the composition method over bounded *tuple* profiles—which is the “correct room” the keystone analysis independently points to (§10).

## 6 Why standard undecidability reductions fail

Undecidability proofs for description logics use a variety of techniques: grid encoding ( $\mathbb{N} \times \mathbb{N}$  domino tiling, requiring functional roles or number restrictions), reduction from the Post Correspondence Problem (PCP), reduction from context-free grammar (CFG) intersection, or simulation of  $S5^3$  via geometric constructions.  $\mathcal{ALCI}_{\text{RCC5}}$  lacks the features that each of these techniques exploits: it has no functional roles, no number restrictions, no arbitrary role box, and no geometric coincidence conditions.

### 6.1 Blocked reduction routes

Table 5 summarizes the blocked routes. The **patchwork property** of RCC5 (Renz & Nebel [2])—local (triple-wise) consistency implies global consistency for atomic networks—actively resists grid

| Source Problem                        | Technique                          | Missing in $\mathcal{ALCI}_{RCC5}$ |
|---------------------------------------|------------------------------------|------------------------------------|
| $\mathbb{N} \times \mathbb{N}$ domino | Graded modalities + transitivity   | Number restrictions                |
| $\mathcal{ALC}_{RA}$ [6]              | Post Correspondence Problem        | Arbitrary role axioms              |
| $\mathcal{ALC}_{RA}^\ominus$ [3]      | Non-empty CFG intersection         | Non-disjoint roles                 |
| $\mathcal{ALCI}_{RCC8}$ (Wessel)      | Grid via TPP/NTPP                  | TPP/NTPP distinction               |
| $L_{RCC8}$ (Lutz–Wolter)              | Linearized topological enumeration | Topological rigidity               |
| $L_{RCC5}(RS^\exists)$                | $S5^3$ via supremum regions        | Supremum closure                   |

Table 5: Why standard undecidability reductions fail for  $\mathcal{ALCI}_{RCC5}$ .

encoding, since tiling reductions require rigid global constraints that go beyond local consistency.

## 6.2 A concrete domino attempt in $\mathcal{ALCI}_{RCC5}$

Can we encode a domino-like grid using PP/PPI chains in RCC5? Consider the concept:

$$\begin{aligned}
X = & \exists \text{PPI}.\exists \text{PPI}.C \sqcap \exists \text{PPI}.\exists \text{PPI}.D \\
& \sqcap \forall \text{PPI}.\forall \text{PPI}.(\forall \text{PO}.(\neg C \sqcap \neg D) \sqcap \forall \text{DR}.(\neg C \sqcap \neg D)) \\
& \qquad \sqcap \forall \text{PPI}.(\neg C \sqcap \neg D) \sqcap \forall \text{PP}.(\neg C \sqcap \neg D)) \\
& \sqcap \forall \text{PPI}.X
\end{aligned}$$

The intent: each  $X$ -node has two PPI-grandchildren colored  $C$  and  $D$  (the “square”), and the universals isolate the colored nodes from all non-EQ neighbors. The recursive conjunct  $\forall \text{PPI}.X$  forces infinite repetition.<sup>3</sup>

**One level is satisfiable.** Without the recursive conjunct,  $X$  has a 3-element model:  $z \text{ PP } y \text{ PP } x$ , with  $z \in C \sqcap D$ . The two grandchildren are forced to be *the same element* (EQ collapse): if  $z_1 \neq z_2$ , they would be related by some  $R \in \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$ , and the universals would require  $z_2 \notin D$  (firing  $z_1$ ’s  $\forall R.\neg C \sqcap \neg D$ ). So  $z_1 = z_2 \in C \sqcap D$ . The  $\forall \text{PP}$  conjunct is *not* vacuous— $z$ ’s neighbors are its PP-successors  $y, x$ , on which it fires—but it is satisfied by interpreting  $y, x$  outside  $C \cup D$ ; the  $\forall \text{PO}/\forall \text{DR}/\forall \text{PPI}$  conjuncts are vacuous, as  $z$  has no such neighbors.

**Two levels is unsatisfiable.** Adding  $\forall \text{PPI}.X$ : the PPI-child  $y$  must itself satisfy  $X$  (without recursion), producing a new  $C \sqcap D$ -labeled element  $w$  as a PPI-child of  $z$ . But  $z$  is a PPI-grandchild of  $x$ , and  $x$ ’s universals fire:  $z$ ’s PPI-neighbor  $w$  must satisfy  $\neg C \sqcap \neg D$ . Yet  $w \in C \sqcap D$ . **Contradiction.**

This was verified computationally: a simplified version  $X_1 = \exists \text{PPI}.\exists \text{PPI}.C \sqcap \forall \text{PPI}^3.\neg C$  is SAT;  $X_2 = X_1 \sqcap \forall \text{PPI}.X_1$  is UNSAT. The domino encoding collapses at depth 2 because the isolation constraints from the grandparent level kill the colored elements created by the recursive unfolding.

**Remark 8.** The failure is not accidental. In  $\mathcal{ALCI}_{RCC5}$ , the only way to create “functional” successors is via PP/PPI chains, but  $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$  makes these chains *transitive*—grandparent universals automatically propagate to grandchildren, preventing the positional diversity needed for grid encoding. In  $\mathcal{ALCI}_{RCC8}$ , the TPP/NTPP distinction breaks this transitivity (a direct TPPI-child is distinguishable from indirect descendants), which is why Wessel’s even-odd chain works there but has no RCC5 analogue.

<sup>3</sup>The recursive occurrence of  $X$  abbreviates a TBox axiom  $X \sqsubseteq F(X)$ ; no fixpoint constructor is needed. Since the five relations partition the square of the domain, the *universal modality*  $[U]D := \forall \text{EQ}.D \sqcap \forall \text{PP}.D \sqcap \forall \text{PPI}.D \sqcap \forall \text{PO}.D \sqcap \forall \text{DR}.D$  is definable, and a finite TBox internalizes:  $C$  is satisfiable w.r.t.  $\mathcal{T}$  iff  $C \sqcap [U]G_{\mathcal{T}}$  is, where  $G_{\mathcal{T}}$  conjoins  $\text{nnf}(\neg C_i) \sqcup D_i$  over the axioms  $C_i \sqsubseteq D_i$  of  $\mathcal{T}$ . The internalization uses  $\forall \text{PO}$ , so it does *not* preserve the  $\forall \text{PO}$ -free fragment of Theorem 10.

### 6.3 A concrete domino attempt in $\mathcal{ALCI}_{\text{RCC8}}$

The remark above suggests the TPPI/NTPPI distinction in  $\mathcal{ALCI}_{\text{RCC8}}$  might rescue the construction: a direct successor (TPPI) and an indirect one (NTPPI) are genuinely distinguishable, so perhaps the level-2 collapse can be avoided. We attempt the analogue and show that while it escapes the RCC5 failure mode, it fails for a different reason—Wessel’s *coincidence obstruction* [9].

**The schema.** Let *Corner* be the concept that isolates a node from all non-EQ neighbors:

$$\text{Corner} = \bigwedge_{R \in \{\text{DC}, \text{EC}, \text{PO}, \text{TPP}, \text{NTPP}, \text{TPPI}, \text{NTPPI}\}} \forall R. (\neg C \sqcap \neg D).$$

The isolation is placed *locally on the colored grandchildren* rather than as an ancestral  $\forall \text{TPPI}. \forall \text{TPPI}$  universal, so it cannot propagate transitively through the recursion:

$$\begin{aligned} X = & \exists \text{TPPI}. \exists \text{TPPI}. (C \sqcap \text{Corner}) \sqcap \exists \text{TPPI}. \exists \text{TPPI}. (D \sqcap \text{Corner}) \\ & \sqcap \forall \text{TPPI}. (\neg C \sqcap \neg D) \sqcap \forall \text{TPPI}. X. \end{aligned}$$

The third conjunct  $\forall \text{TPPI}. (\neg C \sqcap \neg D)$  forces the corner relation  $x \rightarrow z$  to be NTPPI (not TPPI), using  $\text{comp}(\text{TPPI}, \text{TPPI}) = \{\text{TPPI}, \text{NTPPI}\}$ .

**Level 1 is satisfiable, with a real direct/indirect distinction.** Without recursion,  $x$  has TPPI-children  $y_1, y_2$  (the two “direct” positions) and a single corner  $z$  reached by TPPI-TPPI chains from both  $y_1$  and  $y_2$ . The EQ-collapse at level 2 works exactly as in RCC5:  $z_1 \in C \sqcap \text{Corner}$  and  $z_2 \in D \sqcap \text{Corner}$  are mutual non-EQ neighbors only if *Corner* is violated, so under strong-EQ semantics  $z_1 = z_2 = z$ . Crucially,  $z \neq y_1, y_2$  in general:  $z$  is NTPPI of  $x$  while  $y_1, y_2$  are TPPI—the direct/indirect distinction survives, which is a genuine RCC8 refinement over the RCC5 attempt of §6.2.

**Level 2 collapses via the coincidence obstruction.** Adding  $\forall \text{TPPI}. X$ : the TPPI-child  $y$  of  $x$  satisfies  $X$ , producing its own corner  $w \in C \sqcap D \sqcap \text{Corner}$  at NTPPI-distance from  $y$ , and hence at NTPPI-distance from  $x$  via  $\text{comp}(\text{TPPI}, \text{NTPPI}) = \{\text{NTPPI}\}$ . Now  $x$ ’s own corner  $z$  and  $y$ ’s corner  $w$  are distinct grid positions in the intended reading—but as elements of the model they are related by some RCC8 base relation. Since  $z$  carries *Corner*, every non-EQ neighbor of  $z$  lies in  $\neg C \sqcap \neg D$ . But  $w \in C \sqcap D$ . Therefore  $z = w$  under strong-EQ. **All corners across the grid coincide at a single point.** Figure 2 isolates this local “merge attempt” on two successors.

**Why the TPPI/NTPPI trick fails.** In the RCC5 attempt the collapse is a *transitive-propagation* failure: the grandparent universal fires on great-grandchildren because  $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$ . Moving the isolation to a local *Corner* marker and using the non-transitive piece  $\text{comp}(\text{TPPI}, \text{TPPI}) = \{\text{TPPI}, \text{NTPPI}\}$  eliminates that specific pathway—the universal at the grandparent no longer reaches deeper nodes. But a *different* obstruction takes over: the composition table permits any of  $\{\text{PO}, \text{EQ}, \text{TPP}, \text{NTPP}, \text{TPPI}, \text{NTPPI}\}$  between two corners at different intended grid coordinates, and nothing in the concept language rules out EQ. The *Corner* markers themselves then force the EQ-merge. This is exactly the obstruction Wessel identified in report7.pdf, Figure 11 (cf. §4.3): the composition table does not carry the rigidity needed to keep distinct grid cells distinct.

**Remark 9.** The failure mode is structurally different from the RCC5 one but traces back to the same root cause. Lutz–Wolter’s  $L_{\text{RCC8}}$  reduction *circumvents* the coincidence obstruction rather than enforcing it: they linearize the plane into a single NTPP-chain (the Marx–Reynolds enumeration) and recover the vertical direction from the enumeration, so no “north-of-east = east-of-north”

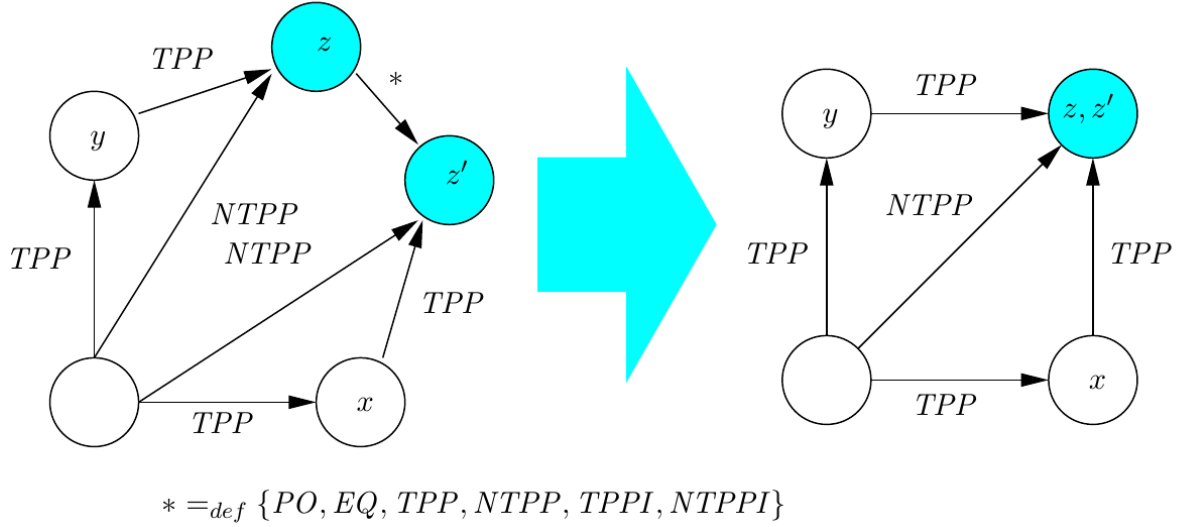


Figure 2: Wessel’s original statement of the coincidence obstruction, reproduced *directly from report7.pdf (2002/2003)*, Figure 11 (“Tightening the grid?”). The two successors  $z$  (reached east-then-north) and  $z'$  (north-then-east) ought to be the *same* grid corner, but the composition table permits the edge  $z * z'$  to be any of  $* =_{\text{def}} \{PO, EQ, TPP, NTPP, TPPI, NTPPI\}$  (the legend below the graphs): the table *by itself* under-determines the edge, forcing neither a merge nor a split. One *can* nonetheless force the merge at the concept level—clash out all five non-EQ options with universals, and EQ is all that remains—and this is exactly the trap: the very **Corner** markers that force  $z = z'$  at one level, once propagated to build the grid, force *every* corner together and the construction collapses (satisfiable at one level, UNSAT at two—§6.2, Remark 2). So neither option yields a rigid grid: leave the edge loose and the branches never join; force the merge and all corners collapse to a single point. It is this “merge the wrong successors” *dichotomy*—not an inability to force EQ at all—that Wessel identified two decades before this AI-assisted work.

coincidence has to be forced; the geometry of  $\mathbb{R}^n$  only has to keep that linear encoding faithful [10]. No such rigidity is available in the abstract RCC8 semantics of  $\mathcal{ALCI}_{\text{RCC8}}$ : a pure composition table allows corner points to merge freely, and the TPPI/NTPPI distinction—while useful locally for a direct/indirect separation at level 2—cannot be chained to enforce an unbounded  $\mathbb{N} \times \mathbb{N}$  grid. This matches the open-problem status of  $\mathcal{ALCI}_{\text{RCC8}}$  decidability noted in [10] and further discussed in [9].

## 7 Overview of approaches, by technique

We organize the attack by *technique*, in roughly the order tried, not by round or date. Each subsection follows the same shape: the high-level idea; the concrete techniques investigated; where and why they fail (with the examples that made the failure clear); pointers to the papers, probes, and reasoners; and the lessons.

### 7.1 Naive tableaux and blocking

**The idea, and why it should work.** The default decision method for a description logic is a *tableau*: to test whether  $C_0$  is satisfiable, try to build a model top-down. Start with one individual

required to satisfy  $C_0$ ; whenever a node must satisfy  $\exists R.D$ , create a fresh  $R$ -successor satisfying  $D$ ; whenever it satisfies  $\forall R.D$ , propagate  $D$  to all its  $R$ -successors; close a branch on a contradiction. Left alone this may never terminate— $\exists PP.\top \sqcap \forall PP.\exists PP.\top$  builds an infinite PP-chain—so one adds *blocking*: if a new node’s “type” (the set of concepts it must satisfy) repeats that of an ancestor, we stop expanding it and reuse the ancestor’s already-built subtree, folding the infinite chain into a finite loop. For ordinary modal and description logics this is sound and complete, and it is the first thing one tries here. Our most refined implementation is the *cover-tree tableau* (§8), which blocks on *exact occurrences*.

**Why the total-relation setting breaks it.** In  $\mathcal{ALCI}_{RCC5}$  every pair of individuals carries a relation, so a model is a complete graph, not a tree. This single fact defeats ordinary blocking in three compounding ways, each of which we hit in practice.

1. **Composition forces edges into already-built nodes.** Creating a new node  $z$  does not just attach it to its parent; by composition it forces relations between  $z$  and *every* existing node. From  $xPPy$  and  $yDRz$  the table forces  $xDRz$  (Proposition 1). So the “tree” is a fiction: the real object is the whole complete graph, and a node’s obligations depend on its relations to the entire context, not on its concept type alone.
2. **Dormant universals reawaken blocked nodes.** A restriction  $\forall R.D$  lies inert at a node until an  $R$ -edge from it appears. But a *forced* composition edge (point 1) can create such an edge *after* the node was blocked, firing the dormant universal, adding  $D$  to a node downstream, and possibly *unblocking* it. The cascade of re-qualifications need not terminate, and—worse—may be *unsound* if ignored: the block asserted a repetition that the later edge invalidates.
3. **A fresh witness’s relations are forced by the joint closure, and can collide with a universal.** Adding  $d$  does not leave its relations to old nodes free: they are pinned by *all* triangles at once, and (Remark 2) the intersection of several can fix even a horizontal edge. Take old  $aDRb$ ,  $aPPIc$ ,  $bDRc$  and a request  $bPOd$ ,  $cPOd$ : this is consistent, but it *forces*  $aPOd$ —from  $\text{comp}(DR, PO) \cap \text{comp}(PPI, PO) = \{DR, PO, PP\} \cap \{PPI, PO\} = \{PO\}$ , an edge  $a$  never chose. Now let  $a$  carry  $\forall PO.\neg A$  while  $d$  must satisfy  $A$ : the *forced*  $aPOd$  fires the universal and closes the branch. That is the real hazard—a value fixed by joint closure, not by any single step, silently contradicting a distant universal (the diagnostic  $C_{\text{force}}$  below is exactly this shape)—and it is invisible to any check that reads edges one path at a time.

**The ladder of blocking conditions we tried.** Each rung tracks more relational context, and each is defeated by the phenomena above. The base choice is *where* one may block: the obvious strengthening of ancestor blocking is *anywhere blocking*—block a node against *any* earlier node in the tableau, not only its ancestors, the more aggressive termination device standard for logics with inverse roles. This helps termination but not soundness: it inherits every failure below, since it still compares nodes by a *local* summary while the constraints are global. On top of the where-question we layered progressively richer *what*-summaries. *Subset/type* blocking (repeat of the concept set) ignores relations entirely and is unsound at once. *Triangle* blocking tracks, for each node, the composition-*triangles* it participates in—the relation triples  $(\rho(x, y), \rho(y, z), \rho(x, z))$ —so that composition constraints are visible; but a triangle is a three-point view, and forced edges from a fourth point (via inverse roles) escape it (`triangle_blocking_ALCIRCC5`, `src/triangle_*.py`). *Quadruple* types lift this to four points to capture those cross-edges (`decidability_via_quadruples_ALCIRCC5`, `src/quadruple_type_check.py`); but the required arity is not bounded—a forced constraint can span a chain of any length. *Profile* blocking records a node’s type together with

the multiset of its relations to a bounded context (`src/profile_blocking_*.py`); but the context that actually determines future obligations is not bounded a priori—this is F6 in disguise. *Double blocking* (the *SHIQ* device: block only when a node *and its predecessor* jointly repeat an ancestor pair, which is what one needs for inverse roles) addresses point 2’s predecessor sensitivity but not the forced-edge and extension problems (`src/double_blocking_test.py`).

**The examples that settle it.** Two families are diagnostic and recur throughout the project. The forced-composition UNSAT concepts, e.g.  $C_{\text{force}} = \exists PP.(\exists DR.A) \sqcap \forall DR.\neg A$ : here  $x PP y$  and  $y DR z$  force  $x DR z$ , so  $\forall DR.\neg A$  kills the required  $z:A$ ; a procedure that ignores forced edges wrongly *accepts* it. Dually, the strong-EQ PO-loop  $C \sqcap \exists PO.\exists PO.C \sqcap \forall \{PO, DR, PP, PPI\}.\neg C$  is satisfiable *only* via a two-element model closed into a symmetric PO-loop under EQ-identity; a procedure that distinguishes occurrences by type alone wrongly *rejects* it (this is precisely the blind spot of the quasimodel oracle, §7.2). Further diagnostics—`inter_witness_counterexample_ALCIRCC5`, `typed_patchwork_counterexample_ALCIRCC5`—exercise the same seams.

**Lesson.** In a total-relation logic, sound termination cannot be had by *local* blocking, because the relational completion is *global*: what may be reused depends on the whole graph, and forced edges plus dormant universals make “the whole graph” irreducible to any fixed-arity local view. The way out is to stop trying to block a complete graph and instead *separate* the tree skeleton from the global relational completion, controlling the latter by the RCC5 patchwork property. That is the split-forest idea (§7.4).

## 7.2 Finite-model and reduction attempts

**The idea.** If one could force models to be finite, or finitely presented, one could enumerate candidates and check them. Four concrete lines: build a *quasimodel* (a set of types with a consistency relation) and eliminate types that cannot be realized; collapse infinite PP/PPI chains into finite *clusters or loops*; take a *two-tier quotient* (one “kernel” per recurrent chain descriptor, plus finitely many designated witnesses); or *encode* satisfiability into monadic second-order logic (MSO) over trees and invoke Rabin’s theorem.

**Where they fail.** The quasimodel / type-elimination procedure (`decidability_ALCIRCC5`, `src/alcircc5_reasoner.py`) is *retracted*: its elimination rule for PO-demands (Q3) is *anti-monotone*—eliminating one type can invalidate the witness that justified another—so the fixpoint over-eliminates and the procedure is *incomplete* (though not unsound). Concretely it is wrong on the PO-loop above: it reports UNSAT for a satisfiable concept. The finite-cluster and MSO routes both stumble on the same rock: collapsing a PP-chain is fine (Lemma 4), but the *horizontal* width across clusters is not bounded, and MSO cannot define an unbounded horizontal matching—so a correct encoding again presupposes a width bound. Every “make it finite” variant reduces to producing a computable live-width bound, i.e. to F6 (Definition 5).

**The win.** One fragment escapes unconditionally. In the two-tier quotient, PO is the *only* relation whose recurrence along a chain needs an “all-phase” safety condition—because, unlike DR/PP/PPI, a PO-witness at one chain position need not remain a PO-witness at another (the compositions  $\text{comp}(PP, PO)$  and  $\text{comp}(PPI, PO)$  are not singletons). If the concept contains *no*  $\forall PO$  subformula, that safety condition is vacuous, the quotient is finite, and the width is bounded by a computable function of the concept.



**Theorem 10** (Decidable fragment). *Concept satisfiability for the  $\forall\text{PO}$ -free fragment of  $\mathcal{ALCI}_{\text{RCC5}}$ —concepts with no  $\forall\text{PO}$  subformula, but retaining  $\exists\text{PO}$ ,  $\forall\text{DR}$ ,  $\exists\text{DR}$ , and all vertical modalities—is decidable, with a computable live-width bound  $K(C_0) = (1 + m)^2 2^{2^m}$ ,  $m = |\text{cl}(C_0)|$ . (`two_tier_quotient_ALCIRCC5`; the closed-form bound is theorem-level, but the underlying `Decidable` instance is machine-checked in `formal/POFreeLift.lean` for the three pure quadrants and the mixed case on a witness—see the 2026-08-06 note below.)*

The fragment is genuinely expressive—one may still say “overlaps some  $A$ ” ( $\exists\text{PO}.A$ ), “all disjoint things are  $B$ ” ( $\forall\text{DR}.B$ ), and impose arbitrary part-of nesting—so this is the project’s strongest unconditional  $\mathcal{ALCI}_{\text{RCC5}}$  result.

**Remark 11** (A gap in the proof of Theorem 10, found and repaired 2026-08-28). The two-tier completeness argument defined the relation between two kernels as a double limit  $\lim_{i,j \rightarrow \infty} \rho(d_i, d_j)$ , justified by its external stabilization lemma “applied twice”. That lemma is one-sided—it fixes an element and varies the chain—and stabilizing every row and every column is strictly weaker than a double limit, which need not exist: in  $\mathbb{Z} \times \{0, 1\}$  ordered by first coordinate, every row and column stabilizes while every tail carries PP, PO and PPI. The step is repaired by replacing the limit with a *simultaneous finite rectangle* (finitely many chains admit segments, beyond any bound, with equal-type endpoints and one constant relation per pair), which is what the quotient needs since it reads only finitely many phases. The repair is machine-checked (`fused_kq_all`). The theorem and its bound are unaffected.

**Re-examined, and its soundness core certified (2026-07-18).** A cold review (§9) rightly flagged that the rationale above leans on the “horizontal freedom” intuition that Remark 2 corrects. We therefore re-examined the fragment; it holds. The soundness crux—the *chain-unfolding lift* (a recurrent chain’s kernel, unfolded under a constant interface, stays a valid RCC5 frame)—is now *kernel-checked* in Lean 4 (`formal/POFreeLift.lean`: `lift_cc`, `unf_is_frame`; axioms `propext/Quot.sound`, zero `sorry`), and it enforces *full* composition-consistency (every triangle closed), which *subsumes* multi-path PO forcing—so the construction does *not* assume PO edges are free, closing the review’s concern. Two independent decision procedures moreover agree on every  $\forall\text{PO}$ -free concept tested (`wp86`, `wp87`: 244 random + curated, zero mismatches; the quasimodel oracle’s lone blind spot requires  $\forall\text{PO}$  and so cannot arise here). A byproduct worth noting: the abstract-semantics core needs *no* patchwork or compactness—a composition-closed, converse-coherent, strong-EQ network already *is* a model, so `unf_is_frame` delivers an actual (infinite) frame outright. What remained for full certification at that point was the model-of- $C_0$  layer (Hintikka demand/universal satisfaction) and the completeness extraction carrying the  $K(C_0)$  bound. So as of that review Theorem 10 stood as *strongly supported with its soundness core machine-certified*—firmer than before, though not yet end-to-end formalized; the following note records how the certification was subsequently carried through to the decision procedure itself.

**Certified as a decision procedure (2026-08-06; superseded 2026-08-29 — see §7.3).** The certification has since been carried through to the decision itself: in `formal/POFreeLift.lean` (then ~13,600 lines, zero `sorry`; axioms `propext/Classical.choice/Quot.sound`, the choice confined to erased termination proofs, so the procedure is genuinely computable) a `Decidable` instance for satisfiability of  $C_0$  is now machine-checked—*generally* for the fragment’s three *pure* quadrants (horizontal, ascending-vertical, descending-vertical) and on a witness for the *mixed* quadrant. Remark 12 records *that* construction: the finite certificate, the four quadrants, and the constructive round-robin uniformization (`rr_covers`) that turns the full logic’s open “W2” oracle into a kernel-checked theorem once  $\forall\text{PO}$  is removed. The mixed quadrant was closed, and the whole fragment

re-certified by a different route, on 2026-08-29 (§7.3); that route has since been through three independent cold reviews. None of it bears on the full logic—F6 is forced *by*  $\forall\text{PO}$ , which the fragment removes.

**Lesson.** Finiteness is the wrong target: models are legitimately infinite (unbounded PP-towers are satisfiable). The right target is a finite *presentation* of a possibly-infinite model.

### 7.3 The $\forall\text{PO}$ -free fragment, certified: the cone scheme (2026-08-29)

This is the route that certifies Theorem 10, and the one to read first. An earlier route, recorded in Remark 12 below, certified three of the fragment’s four quadrants generally and the fourth only on a witness; its general mixed extraction was never completed, because two rounds of cold attack refuted the architecture it needed (a finite node set reused by blocking and *borrowing* witnesses) by exact finite countermodels. It was replaced.

The route now certified is a *cone scheme*, designed by GPT-5.6 Sol [62]: a finite control graph whose states are pairs (support type, upper bound on the types of strict PP-predecessors); a monotone, reductive elimination to a greatest fixed point; and a *fresh-occurrence* unfolding in which no element is ever reused, so that the frame conditions which defeated the previous architecture hold structurally rather than by construction.

The *completeness* direction deserves its own sentence, because it is where this route departs furthest from everything before it. Earlier architectures read completeness as *extraction*: given a model, build a finite certificate out of it. That construction is where the width budgets live, and where four successive disciplines were refuted. Here nothing is extracted. The signature space is fixed in advance from  $C_0$  alone; a model point’s signature is *read off* it — its type, together with the types of its strict PP-predecessors; and the set of signatures a model happens to realise is shown to survive its own elimination, because for each demand the model simply hands over a witness. That makes it a *post-fixed point*, and since the elimination is monotone and reductive, any post-fixed point inside the starting set lies inside the greatest fixed point. Finiteness costs nothing — the signature space was finite before any model was mentioned — and the model’s entire contribution is that one lemma.

This is also exactly the trap the retracted quasimodel route fell into (§7.2): its elimination was *anti-monotone*, so no such argument was available to it and its fixpoint over-eliminated. The elimination here asks only that each demand *have* a target in the current set, which is monotone — and that difference is what makes the greatest-fixed-point argument possible at all. It was checked before anything was built on it, precisely because this project had made the opposite mistake once.

The design also explains the fragment boundary better than the syntactic statement does, and the explanation is worth recording because it says what  $\forall\text{PO}$ -freeness is *for*. A composition cell containing PO can always be discharged by putting the pair at PO — unless some formula can object to a PO edge, and the only formula that can is  $\forall\text{PO}.D$ . So on this fragment the maximally-PO unfolding, which declares only what a birth forces and defaults every remaining pair to PO, is *always* a model; no joint realizability across a signature’s several demands is ever required, because every cross-relation between distinct children is the residual and the residual costs nothing. What is lost when  $\forall\text{PO}$  returns is therefore not one case of one induction but the residual default itself — which is the same boundary Remark 2 locates from the side of joint forcing. The capstone is

$$\text{decidableSat\_cone} : \text{POFree } C_0 \rightarrow \text{Decidable}(\text{Satisfiable } C_0),$$

with  $\text{POFree}$  the only hypothesis, covering all four quadrants. Its completeness half needs no fragment hypothesis at all, which yields a sound UNSAT test for the *full* logic. That test is one-sided

and, in its first form, was also *empty*: a second cold review measured its verdict invariant under deleting every  $\forall\text{PO}$  from the input, and that deletion always lands in the fragment, so it refuted nothing the fragment procedure did not already refute. The cause was a dropped clause —  $\text{PO}$  is its own converse, so  $\forall\text{PO}$  bodies cross a  $\text{PO}$  edge in both directions — now restored, with a machine-checked concept that the repaired test refutes and whose deletion is satisfiable. A second  $\forall\text{PO}$  gap remains and is architectural: the scheme exploits only compositions forced to a singleton, so obligations arising from disjunctive entailments are out of its reach.

Two scope limits recorded in earlier versions of this remark have since been closed. The artifact now carries a *raw* syntax with negation, a normalization, and a preservation theorem in both polarities, so no pre-normalized input is assumed (`decidableFSat`); the polarity flag also exposes that the fragment condition on raw input is *no  $\forall\text{PO}$  positively and no  $\exists\text{PO}$  negatively*, since an  $\exists\text{PO}$  under a negation becomes a  $\forall\text{PO}$ . And satisfiability over the abstract composition-table semantics is proved *equivalent* to satisfiability over families of non-empty sets under the naive relation — equal, proper subset, proper superset, disjoint, otherwise overlap — with no fragment hypothesis, so the equivalence covers the whole logic (`satisfiable_iff_set`); on that side the composition table is a theorem about sets rather than an assumption. Regions there are arbitrary non-empty sets, not regular closed sets in a topological space.

What remains is a limit of complexity, not of correctness: the procedure is *decidable but not runnable*. The signature space is doubly exponential, and the procedure cannot be evaluated on any concept discussed in this paper.

Three cold reviews (all 2026-08-29) built the artifact, re-derived the composition table independently, and none found a counterexample. The first cross-checked 4,000 random concepts against exhaustive finite-model search and found two inverted comments and a complexity bound understated by an exponential. The second attacked completeness and *ran* the elimination — 6,268 concepts against exhaustive model search, no satisfiable concept rejected — and found the dropped  $\text{PO}$  clause above, together with an overclaim about what the full-logic test certifies. The third attacked soundness, ran the unfolding on labels drawn from the actual greatest fixed point, and found no defect in the development: its five findings were all in the project’s own probes and prose. All are corrected.

The honest label, and it differs from this paper’s standing one: the  $\forall\text{PO}$ -free fragment is *machine-certified* — the Lean kernel checks the decision procedure with fragment membership as its only hypothesis — and its formalization’s adequacy has been examined by three independent cold reviews without a counterexample. The project’s blanket *strongly supported, not certified* applies to the *full logic*, and to the two unformalized arguments of §7.2; it does not apply to this result, and earlier drafts of this section were wrong to carry it over.

**Remark 12** (The superseded first route: four quadrants and a round-robin kernel). Everything here is in the artifact and every theorem named is still proved; what changed is which construction discharges Theorem 10. This route certified a `Decidable` instance *per quadrant*, generally for the three pure ones and only on a witness for the mixed one; its general mixed extraction was never completed, and two rounds of cold attack later refuted the architecture it needed. The route that certifies the theorem is the cone scheme of §7.3, which uses none of the machinery below.

It is recorded because the material is not superseded as *mathematics*: the four-quadrant taxonomy is a real classification of how demands interact in the fragment, and the round-robin device answers the uniformization question the full logic leaves open. It is also the clearest place to see *why* the fragment is tractable at all.

**The certificate, and the decision loop.** A certificate is a *multi-tier* object (`MultiTier`): finitely many *external* neighbour-types together with finitely many recurrent *kernels*—chain gener-

ators, read as “the pattern continues like this” and unfolded into infinitely many distinct elements, never bent into a finite quotient. Each is decorated with Hintikka bookkeeping: which existential demand each node discharges, which universal obligations propagate along which edges, and a declared RCC5 value for every relevant pair, required composition-closed. A total Boolean checker `mtAcceptB` accepts a certificate exactly when all of this holds, and acceptance is *sound*—a passing certificate unfolds, through a Hintikka labelling and truth lemma, to a genuine RCC5 model of  $C_0$  (`mtAcceptB_sound`). Certificates are finite list data with a total Gödel decoder (`FinMT`), so the decision procedure is immediate (`decidableSat_of_codes`): enumerate the finitely many codes a concept admits, run the checker, answer SAT iff one passes. All the mathematics is in emitting the right code for a satisfiable concept; the algorithm itself is a bounded for-loop.

**Four quadrants.** A  $\forall$ PO-free concept’s existential demands fall into four combinations, each with its own certificate shape:

- *Horizontal* ( $\exists$ DR,  $\exists$ PO,  $\exists$ EQ only): a node’s label is its model type *truncated at the concept’s modal depth* (`mtk`). The frame is universally closed in both directions (disjointness is self-converse, so a  $\forall$ DR obligation propagates identically each way), and depth-truncation terminates the recursion, so finitely many labels exhaust the demands (`decidableSat_hfrag`). No kernel is needed—the horizontal world is bounded outright.
- *Ascending vertical* ( $\exists$ PP): a single recurrent kernel generating one PP-chain. Along any ascending chain the Hintikka type recurs with a *bounded period* (an infinite-pigeonhole segment lemma, `mty_segment_bounded`); that period *is* the kernel (`decidableSat_vtower`).
- *Descending vertical* ( $\exists$ PP $\downarrow$ ): the exact dual. The certificate carries the chain’s direction as a single bit, so the whole encoding and decision side is direction-agnostic and the ascending development transports verbatim (`decidableSat_vtowerRRI`).
- *Mixed* ( $\exists$ PO and  $\exists$ PP): the merged certificate `mixCert_ok`—the first that carries *both* non-trivial externals *and* a kernel. Each  $\exists$ PO demand is served by a direct external neighbour, each  $\exists$ PP demand by the kernel, and every cross-obligation living on a PO-edge is *vacuous*, because no  $\forall$ PO clause occurs anywhere in the fragment’s closure (`mty_no_all_po`). Its encoding and a complete decision are certified on a concrete witness (`decidableSat_Cmix`).

**This route’s keystone: a constructive uniformization.** (The cone scheme does not need it; its own keystone is the residual-PO default of §7.3.) One recurrent kernel must often serve *several* distinct  $\exists$ PP demands at once, and on the full logic this is exactly the feared uniformization property  $W2'$ : can a single repeating interface discharge every demand without two of them colliding on a horizontal edge? In the fragment the answer is *constructive*. The kernel serves the demands in *round-robin* over its recurrent period—at period position  $n$  it discharges the  $n$ -th demand, drawing the witness from whatever phase carries it—so across one full period *every* demand is met (`rr_covers`). This turns  $W2'$  from an oracle into a kernel-checked theorem, and it works for one precise reason: with  $\forall$ PO absent, the horizontal edges the kernel emits at successive steps are *unconstrained*, so the phases never conflict and the round-robin schedule is always realizable. That is the exact sense in which removing  $\forall$ PO lets the cross-relations coordinate for free.

**Why the boundary is exactly here, and the honest labels.** The construction stops precisely at  $\forall$ PO-freeness because F6—the full logic’s open live-width keystone—is *forced by*  $\forall$ PO: it is a  $\forall$ PO obligation that can watch a kernel’s emitted horizontal edges and so demand a bounded, coordinated interface across phases. Remove that one constructor and the coordination F6 abstracts becomes free, which is what every device above cashes in. Scope of *this route*, stated exactly: the three *pure* quadrants are certified *generally*, each with a non-vacuous satisfiable witness (`Cvert`,

Ccar, Calt, Clin, Cdesc); the *mixed* quadrant’s certificate, encoding, and decision are certified only on a witness. Its general extraction was never completed; §7.3 above is what replaced it.

*Correction (2026-08-28).* An earlier version of this sentence called that extraction “scoped, not open”. That was an overclaim and is withdrawn: two rounds of cold attack have since refuted the extraction architecture it referred to—a finite node set reused by blocking and borrowing witnesses—by exact finite countermodels, and the formalization has been restarted on a different architecture (a finite control graph with a fresh-occurrence unfolding, in which no node is reused). This concerns the *certification* of Theorem 10, not the theorem, whose proof is the two-tier quotient and is independent of that route; see Remark 11. The certified route is no longer unreviewed: three independent cold reviews (2026-08-29) attacked its trusted base, its completeness direction and its soundness direction, and none found a counterexample or a defect in the fragment result. The two *arguments* of Theorem 10 remain unformalized, and the standing ledger presumption (§9) continues to apply to them, and to every full-logic claim in this paper.

**Why it failed, and what that generalizes to.** The four disciplines tried in succession were not four unrelated mistakes; they were four attempts at one impossible thing. Reusing a node imposes a *global* condition on the object being built — the order induced by the reused edges must stay acyclic, and the relations those edges force must be *jointly* realizable — while every discipline proposed was a *local* test: per edge, per node, per type. A local test cannot see what it is not looking at, and the refutations say so one axis at a time. Per-edge safety removes self-loops and nothing longer: the countermodel has two individually safe fallbacks forming a four-cycle. A key computed from an occurrence’s downward cone cannot separate occurrences whose obstruction is *horizontal*. A class top defined by type alone drags one member’s disjointness witness onto another member’s cone. This is the same pointwise-versus-joint shape that had already broken three earlier architectures here, which is why the attempt was abandoned rather than repaired: the error was proposing another local discipline, not the particular disciplines proposed. Standard *blocking* is this story in its most familiar form — it is precisely the local test, “two occurrences carry the same type, so let one stand for the other” — and it fails here for exactly the same reason: type equality cannot see whether the reuse closes a cycle in the order, or places the blocker’s disjointness witness inside the blocked node’s cone, which is the clause (1tNotDj) that broke this project’s last and most careful attempt at it.

What §7.3 does instead is not a better test — it removes the obligation. Nothing is reused, so the order is a tree closure, irreflexive because paths cannot return rather than because a check forbids it; and a DR birth opens a new vertical component, so the order and the disjointness relation can never meet. The global conditions hold *structurally*. There is no test to get right, because there is no way to violate them. That is the general lesson worth carrying out of this project: when a construction keeps failing a global consistency condition, the productive move is not a sharper local check but a shape in which the condition cannot be expressed as a failure.

## 7.4 The split-forest idea

**Idea (Wessel’s, on a whiteboard).** The split-forest decomposition was Wessel’s contribution, sketched on a whiteboard (Figure 3) and handed to the AI assistants to make precise. The insight: even though a model is a complete graph, one can *present* it as a *forest of trees*—one tree per connected component of the vertical (PP) order—by *splitting* each node that is a common part of several superparts into separate *occurrence copies* (which are *not* EQ-related model elements—under strong EQ that would mean identity—but occurrences of an occurrence cover, re-identified afterwards by an explicit quotient), so the tree skeleton becomes explicit, and then *completing* the missing (horizontal) edges by the RCC5 patchwork property. Tree-like presentations of infinite



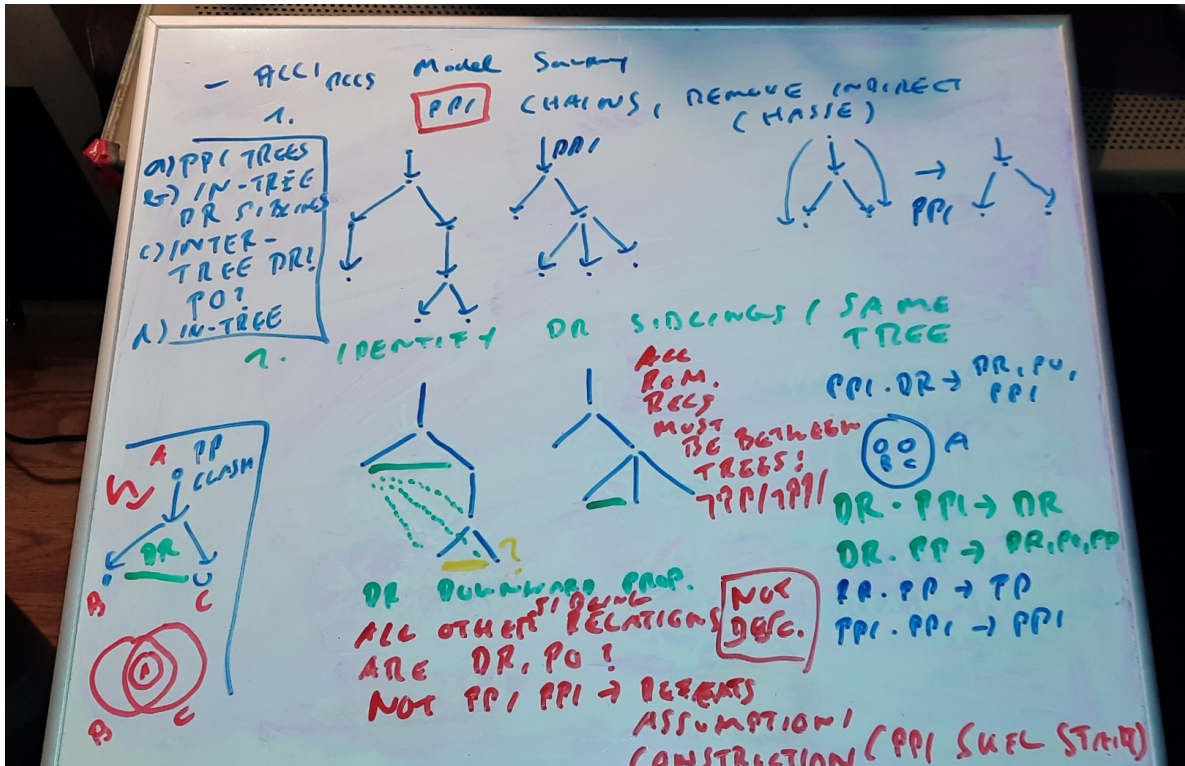


Figure 3: The original whiteboard sketch of the split-forest model decomposition (M. Wessel, April 2026): the intuition that a complete-graph RCC5 model can be presented as a forest of trees by splitting join nodes into occurrence copies (re-identified by a quotient) and completing the relational graph by patchwork. This idea—Wessel’s—is the semantic foundation that every subsequent proof route consumes, and the one component no cold review ever broke.

models are the classical gateway to decidability: once a logic has a tree-model property, automata or type-elimination usually finish the job. The bet was that this setting, despite its complete relational graph, still admits such a presentation.

**The normal form (Theorem A).** Made precise, the construction is a chain of thinning and splitting operations that every satisfiable concept survives: *witness thinning* (keep one witness per demand) → *generated containment covers* (organize witnesses by the vertical order) → *split copies* for incomparable superparts → *side-interface mosaics* (the horizontal relations at each interface, completed by patchwork) → a *typed-EQ quotient* (re-identify the split mates by exact occurrence). The output is a witness-generated, tree-like split-forest model. This normal form is the one component found sound by *every* cold review across the project—the semantic foundation everything downstream *consumes* rather than replaces (`trees/`, `cover_tree_tableau_ALCIRCC5`).

**Vertical skeleton, horizontal completion.** The presentation enforces a strict division of labour between the two axes, and that is what makes it a *forest* rather than an arbitrary graph. Every *tree edge* is vertical: a parent-to-child link is a PP/PPI (part-of) step, so each tree is literally a branch of the containment order. Every *non-tree edge*—between siblings, cousins, or nodes in different trees—is *horizontal* (DR or PO). A genuine vertical relation that happens to hold between two nodes not yet on one ancestor line is never left lying across the forest: it is *re-oriented* into the tree



(the smaller region is made a descendant of the larger), so that once the splitting is finished the only edges the trees do *not* already display are horizontal ones. This is exactly why the completion problem is a purely *horizontal* one—and why PP/PPI never appear as edges to be guessed at an interface.

Half-determinism (Proposition 3) then does most of the completion for free. Once a DR edge is placed between two subtrees, *every* pair of descendants across them is forced to DR—all shadows, none of them live—so a DR crowd of any width is generated by a few high DR edges and pushed downward deterministically, costing a finite presentation almost nothing. What resists this is precisely the residue: PO (never forced by a single step, and not downward-propagating, Proposition 1) and the *upward* direction of disjointness (a superpart of a disjoint region is unconstrained). The genuine branching a finite presentation must record—the live width (Definition 5)—is this horizontal residue and nothing else, which is why F6 is a statement about horizontal crowds alone.

**Why it does not, by itself, decide.** The split forest gives the *shape* of models; it does not give a terminating *search*. Turning “a tree-like model of this shape exists” into “an algorithm can decide whether one exists” requires a *finite abstraction* of the (still infinite) split forest—and that finite-abstraction problem is exactly where the next two families live, and where the difficulty concentrates.

**What survived, and how (a reflection, 2026-08).** It is worth recording which parts of the whiteboard idea are still standing in the one place where a decision procedure was actually certified (§7.3), because the answer is unusually clean: *the shape survived and almost none of the machinery did*.

The certified model is the fresh-occurrence unfolding, and it *is* a split forest, in a fairly literal sense. Its elements are demand words rather than nodes, so two occurrences of the same type are different elements—that is the splitting, and this route never undoes it. Its order relation is generated by exactly two moves, “prepend a PP birth” and “strip a PPI head,” so the skeleton is purely vertical; the horizontal relations are not part of it but are *read off* afterwards, DR from the downward closure of the DR births and PO as the residual. And the forest is genuinely a forest: a function stripping leading vertical births returns each occurrence’s component root, an order edge provably preserves it while a DR birth provably changes it. One tree per connected component of the PP order, cross-linked horizontally: the sketch, unchanged.

*Where in the certification each half of the idea is used*, since the two directions use it very differently. On the **soundness** side the split forest is what the procedure *builds*: `coneScheme_sound` hands the truth lemma the fresh-occurrence unfolding, and that object is a split forest in the literal sense above — vertical skeleton, one tree per component of the PP order, horizontal relations read off afterwards. Every model the procedure ever exhibits is of that shape.

On the **completeness** side it is *not* a hypothesis at all. `coneScheme_complete` takes an arbitrary `RCC5Interp` — any model whatsoever, tree-presented or not — and this matters, because a completeness proof that required split-forest models would have to produce one, which is exactly the extraction step that four architectures died on. What survives there is the idea’s *organizing principle* rather than its shape: a point is summarized by its own type together with the types lying strictly below it in the PP order (`sigCone` = support type, then down-spectrum), which is the whiteboard’s claim that vertical position is what characterizes a node — now read off any model rather than imposed on it. So the split forest is where the certified procedure’s models *live*; it is not a condition the certification asks of the input.

That is worth spelling out, because “not a hypothesis” reads easily as “not used,” and here it is

the opposite. The idea does three jobs in the certified development. It is the shape the soundness side *constructs*, and the two frame conditions that defeated the previous architecture — irreflexivity of the order, and its disjointness from the disjointness relation — hold structurally only because that shape is a forest of vertical trees: an order edge cannot leave a component, and a DR birth opens a new one. It is the content of the normal form (Theorem 15), which is what replaced the patchwork appeal and makes composition closure free. And it fixed what a signature records — vertical position, a type together with the types below it — which is why the transition conditions push cones along PP and PPI and ask nothing of PO. What changed is the idea’s *job*, not its importance: the earlier routes needed it as a property to be *established* of an arbitrary model, which is the extraction problem they were refuted on, whereas here it is a shape to be *built*, which costs nothing.

What did *not* survive is everything built to construct and control such a forest. There is no patchwork appeal—composition closure is derived from the ordered-disjoint axioms (Theorem 15), which is a certified general fact about RCC5 rather than a citation. There is no quotient—the whiteboard description ends “re-identified afterwards by an explicit quotient,” and that step simply never happens, because nothing needs to be recovered: any model will do. And there is no witness thinning, no generated containment cover, no side-interface mosaic and no  $K(C_0)$  width accounting; the finite object is the control graph of signatures, and the width question never arises.

The honest reading is not that the machinery was wasted. It is that three successive architectures were each refuted at the point where they tried to *build* a finite presentation of such a forest and keep it consistent, and the move that finally worked was to stop building one—to let the syntax’s own demand words be the forest, and check the labels against it. That move is available only once  $\forall PO$  is removed, for the reason §7.3 gives: only then is the residual default free. For the full logic the finite-abstraction problem of the preceding paragraph is still there, and F6 is still exactly its residue. What the fragment shows is that when the abstraction problem *dissolves*, what is left is the whiteboard picture and nothing else.

## 7.5 The no-automata certificate, and the automata route

**The hand certificate (rounds 2–10).** The first attempt at a finite abstraction was a hand-built *certificate*: a finite object whose validity (checked by local conditions on converse, composition, universal safety, request discharge, and residual frontiers) was meant to be equivalent to satisfiability (gpt5.5\_round2/, gpt5.5\_round4/). Across ten rounds it worked only by importing, piece by piece, machinery that is really *automaton-shaped*: the conditions that keep a certificate finite while a model is infinite are “request-closed cycles” (a demand and its witness recur), which behave exactly like Büchi/parity acceptance, and “finite product-state exhaustion,” which behaves like reachability in a product automaton. Cold reviews found a succession of gaps—an infinite ancestor PPI-tower the construction missed (D-1), then its dual descendant PP-tower, then boundary non-determinacy, then (twice) that “finite checkability” was asserted rather than proved. The honest reading is that the hand certificate was an automaton written informally; so the project made the automaton explicit.

**The parity-automaton route (rounds 11–12).** This organizes the argument into three theorems. **Theorem A:** the split-forest normal form (above). **Theorem B:** a *finite abstraction* (frontier/pair/triple summaries of the split forest) captures saturated split forests up to satisfiability of all closure formulas. **Theorem C:** a two-way alternating parity *tree* automaton accepts exactly the valid abstractions, and its emptiness is decidable (Vardi). The automaton *consumes* the split forest; it does not replace it. Theorem B is the single load-bearing keystone. Cold reviews (the 5th and 6th) found its global-coherence step *assumed, not proved*: it derived a globally consistent

RCC5 labelling of an infinite tree from purely local checks, via a “confluence law” that was never verified. Round 12 rebuilt the abstraction on *patchwork bags* (finite complete atomic networks that agree on overlaps), so global consistency follows from the citable RCC5 *patchwork property* (Renz–Nebel), and the countable completion by a compactness argument over the finite relation alphabet; eventualities are decided by two-way emptiness. A 7th (hot) review then found a residual *Shadow Amalgamation* gap in the truth lemma: the declared shadow edges were asserted to be jointly realizable with the bag networks, which is false without a further discipline (`automata_route_repairs/`, `cold_review_theoremB/`, probes `wp13–wp17`).

**Why we abandoned the tree automaton.** A cold review of the decision layer exposed a structural problem. The automaton reads a labelled tree and decides accept or reject. But the tree does *not* carry all the data that makes an abstraction valid: it omits the relations between each fresh witness and the “sources” of the universal restrictions ( $\forall R.D$ ) that act on it. A tree is valid exactly when *some* consistent choice of those missing relations exists—so the automaton would have to solve a *guessing* problem (“does some completion of the missing data work?”), not merely read off the tree. The trouble is how an alternating tree automaton works: it explores the tree in many independent parallel copies, and each copy would guess the missing relations *locally*, with nothing forcing different copies to guess *consistently*. One can therefore build a tree in which two copies, running over two different witnesses, each privately guess relations that are locally fine but mutually incompatible, so every copy accepts—even though the true relation between those two witnesses is forced by composition to be impossible ( $\text{comp}(\text{PP}, \text{PP}) \cap \text{comp}(\text{PP}, \text{DR}) = \{\text{PP}\} \cap \{\text{DR}\} = \emptyset$ ). The automaton accepts an unsatisfiable input: acceptance without validity. The three natural repairs (annotate the tape—but the enrichment is unboundedly large; quotient the profiles—provably lossy; guess canonically—needs an unproven uniformization) are each blocked by facts the construction itself establishes. So the tree automaton was replaced by a mosaic / type-elimination procedure directly on the infinitary abstraction (§7.6). The keystone survived the change untouched: Theorem B’s finite-abstraction adequacy is F6 in automaton clothing.

## 7.6 The Lean line and F6

**The mathematical basis, and the pivot that produced it.** The Lean development is *not* an automaton—and that is the point of a deliberate pivot. The route of §7.5 tried to decide satisfiability with a two-way parity *tree automaton* (Theorem C); the cold review that found “acceptance without validity” showed that a tree automaton cannot work here, because its independent parallel run copies cannot be forced to guess the missing horizontal relations *consistently*. Rather than patch the automaton, the project moved the decision *off* it and back onto the semantic objects the split forest already supplies. What resulted is a *cluster-quasimodel* construction in the *mosaic / type-elimination* tradition of modal and description logic: a finite catalogue of interface *clusters* (mosaics), a consistency relation for gluing them (controlled by the RCC5 patchwork property), and a *Hintikka truth lemma* for correctness—the same tableau-family argument as §7.1, but run on a *finite abstraction* rather than on an unbounded tableau. Unlike the retracted early quasimodel (§7.2), the procedure *enumerates and finitely checks* catalogues rather than running an elimination fixpoint, so it does not inherit that route’s anti-monotone incompleteness. The split-forest normal form (Theorem A) is the shared semantic foundation that both the automata route and this one *consume*.

What makes the automaton *unnecessary*, rather than merely replaced, is *one-step demand fulfilment*. An automaton earns its keep on *eventualities*: promises—“some part satisfies  $D$ ”—that may be discharged arbitrarily far away, and whose fulfilment along an infinite branch is exactly

what a Büchi/parity acceptance condition checks. Here every  $\exists R.D$  is instead fulfilled by a witness *already present in the catalogue*, so there are no promissory obligations, no eventualities, and hence no acceptance condition to decide—the one task only an automaton did well. Decidability then reduces to enumerating finite catalogues and checking each by a finite composition test, with no automaton in the loop. This is the construction that was formalized in Lean over rounds 19–30.

**Lean, and how we use it.** Lean 4 is a proof assistant whose kernel type-checks every inference and refuses a proof with a gap. In a project where prose proofs repeatedly *assumed* the hard step, the kernel is a ground-truth oracle that cannot be talked past. We moved the load-bearing soundness argument into *Lean 4 core*—no `mathlib`, no external libraries, no build system: every object is built from first principles, which keeps the trusted base small and the axiom footprint auditable (only `propext` and `Quot.sound`, plus `native_decide` in isolated witness lemmas). The encoding is deliberately direct, so that the Lean statements are legible as the mathematics they formalize: the five RCC5 atoms are an inductive type with the composition table and converse as total functions; concepts are an inductive type in negation normal form (inverse roles absorbed via the converse); an *interpretation* is a domain predicate together with a *total* relation function  $\rho : \Delta \times \Delta \rightarrow \{\text{EQ, PP, PPI, PO, DR}\}$  and an atomic valuation; satisfaction is defined by structural recursion (a role  $R$  is read directly as  $\rho xy = R$ ); and the RCC5 frame conditions (reflexive-EQ, strong-EQ, converse-coherence, composition closure) are a **structure** over  $\rho$ . On top of this foundation sit Hintikka labellings and the certificate objects described below. The development is `formal/Round19Transport.lean` ( $\approx 5,300$  lines, zero `sorry`), with the full artifact-by-artifact history in `LEAN.md`.

**What a certificate is.** Concretely, a *certificate* is an explicit finite object whose validity is checkable by finite composition tests. It is worth spelling out what one looks like, since “catalogue” and “steering” recur throughout. A certificate has two parts.

- A *catalogue*—the finite blueprint. It consists of: a *core network* (a fixed finite complete atomic RCC5 network—the part of the model that does not repeat); a finite set of *attachment templates* (each template is one recipe for gluing a fresh cluster of witnesses onto an existing interface—which old occurrences the new ones attach to, in a *gluing normal form* that keeps the shared part canonical); and a *steering function*, indexed by template, that fixes the *horizontal* relations from the old occurrences to each fresh one. The steering function is the one genuine degree of freedom, because those are exactly the relations composition does not determine (Proposition 1); the vertical edges are already carried by the tree.
- A *plan*: a finite recipe naming which template to apply at which interface, in order. *Unfolding* the plan—applying templates step by step, possibly reusing one template unboundedly often to grow an infinite PP-tower—produces the (possibly infinite) split-forest presentation of §7.4.

A certificate is *sound* if a finite check on the catalogue alone—the “S-condition,” independent of how long any plan runs—guarantees that *every* unfolding is a proper, composition-closed RCC5 network; this is the step certified in Lean (below). Its *live width* is the largest number of live (non-shadow) occurrences that must coexist at any one interface during unfolding—the quantity F6 must bound. Demand fulfilment is *one-step*: every  $\exists R.D$  is satisfied by a witness actually present in the catalogue, so there are no promissory obligations and hence no parity/eventuality condition of the kind the tree automaton needed.

It was on this construction, finally explicit and free of hidden choices, that the entire remaining difficulty localized to two named properties: **F6** (Definition 5, a computable live-width bound) and a uniformization property **W2'**.

**Definition 13** (Uniformization, W2'). Extraction reads a finite certificate off a model by summarizing each interface into a bounded *state* (a type, plus the relations to the finitely many recorded anchors). W2' asks that this summary be *faithful*: whenever two old occurrences carry the *same* state, a fresh witness can be glued onto both in the *same* way—so a glue step depends only on the state, and the catalogue's finitely many states really do capture every gluing. If W2' fails, two occurrences that look identical to the catalogue may nonetheless demand different completions, so the finite catalogue *under*-represents the model and the checker can *reject a satisfiable concept*. The error is one-directional: a failure of W2' can only cause over-rejection, never acceptance of an unsatisfiable concept—so W2' bears on completeness alone, never on soundness.

**What is machine-checked (rounds 19–30).** The soundness pipeline is kernel-checked end to end: certificate syntax  $\rightarrow$  an S-condition on the certificate  $\rightarrow$  a proper atomic RCC5 frame  $\rightarrow$  the logic layer (a truth lemma, with one-step fulfilment)  $\rightarrow$  an actual RCC5 model of  $C_0$ . So is the *faithfulness* of the finite (Hintikka) abstraction: satisfiability and abstraction-realizability coincide. On top of these sits a fixed, non-oracular Boolean checker of first-order finite certificate data, proved sound, from which decidability follows by finite search—modulo exactly one premise,  $F6 \wedge W2'$ .

**Theorem 14** (Conditional decidability). *If F6 holds—the live width of models of every satisfiable  $C_0$  is bounded by a computable function of  $C_0$ —then concept satisfiability for  $\mathcal{ALCI}_{\text{RCC5}}$  is decidable. The soundness half of this reduction (a valid finite certificate provably unfolds to a genuine RCC5 model of  $C_0$ ) and the faithfulness of the Hintikka abstraction are machine-checked in Lean 4 (zero sorry). (The uniformization W2' folds into F6—see below—so the sole open premise is F6.)*

**The keystone, sharpened.** Two cold attacks on F6 turned the vague “bounded width” into a precise dichotomy. The naive ways to force width—shell half-graphs, prefix pumps, single-chain PO-gaps—all *collapse* to finite catalogues (they have  $O(1)$  separator cover, a carried memory slot suffices). The sharp obstruction is an unbounded family of  $M_n$  *identity-selector minors*: pairs  $(L_i, U_i)$  each separated by a *private* selector  $A_i$  and by no other, so the selector incidence matrix is the  $n \times n$  identity—private matching of unbounded hitting number. In plain terms:  $n$  separate “this-region-not-that-one” witnesses, no two of which may be merged because each is the *unique* reason some specific pair stays apart—so a finite catalogue, which can only hold boundedly many such witnesses, is overrun exactly when a concept forces  $n$  of them for every  $n$ . F6 fails iff some concept *forces* such minors; F6 holds iff none does. And W2' (Definition 13), which looked like a second obstacle, was shown false in its coarse form but repairable by a state refinement whose size is bounded *by F6*—so it *folds into* F6, leaving a single open question (`fable5_width_barrier/`, `cold_review_f6_w2prime/`; probes `wp35–wp44`, incl. `wp44`: arbitrary RCC5 glue steps *can* have unbounded separator cover, so any F6 proof must use  $\mathcal{ALCI}_{\text{RCC5}}$ -forceability, not RCC5-consistency alone).

**The problem is  $\Pi_1^0$ , and F6 weakens to its qualitative core.** One further observation—new to this project—sharpens what F6 must deliver. As §5 sets up, the semantics is finitely *first-order* axiomatizable (the theory  $T_{\text{RCC5}}$  plus the standard translation of  $C_0$ ), over *arbitrary* carriers—exactly what the carrier-polymorphic satisfiability of the Lean development means.



Concretely this is a *computable reduction to plain first-order satisfiability*: each  $C_0$  maps to the single sentence  $\bigwedge T_{\text{RCC5}} \wedge \exists x \text{ST}(C_0, x)$ , which has a first-order model iff  $C_0$  is satisfiable. And first-order satisfiability is itself  $\Pi_1^0$ —the co-r.e. side of Church’s undecidability theorem, which is Gödel completeness in action: unsatisfiability equals refutability, and FO refutations enumerate. So satisfiability is *in*  $\Pi_1^0$ —and this is an upper bound only: we establish *membership*, not  $\Pi_1^0$ -hardness (a *decidable* problem is also in  $\Pi_1^0$ ). It shares its *ceiling* with the domino problem, but we do not show it as hard, so the “knife’s edge” (§1.3) is a suggestive analogy here, not a completeness result. The *reduction* alone would not even fix this ceiling: a non-first-order model condition—well-foundedness, finiteness, anything genuinely second-order—could push it higher; the finite axiomatization is what keeps the reduction a single sentence.

The consequence for the keystone is best seen through the missing *dual*. A  $\Pi_1^0$  problem is decidable iff it is *also* recursively enumerable, and completeness hands us one such side—enumerability of UNSAT—for free. The classical way to get the *other* side, enumerability of SAT, is the *finite model property* (enumerate finite models), the textbook route “recursively axiomatizable + finite model property  $\Rightarrow$  decidable”—a principle report7 itself already invoked, en route to a bounded variant of these logics [9]. But  $\mathcal{ALCI}_{\text{RCC5}}$  has *no* finite model property (§5: the ascending PP-tower needs an infinite model), so that route is shut, and enumerating SAT the honest way means enumerating *finite certificates for possibly-infinite models*—which, for *this* certificate scheme, is *qualitative* F6: every satisfiable concept has *some* finite certificate accepted by our checker, *with no computable bound demanded*. One dovetails such a certificate enumeration (its checker the already-certified `finAccept_sound`) against the enumeration of FO refutations; on every input, one side must eventually fire. This is a *sufficient* route, not the recursion-theoretic minimum: decidability is equivalent only to the existence of *some* sound, decidable, complete certificate relation, and completeness of *this particular* split-forest/live-width checker is not known to be necessary. So F6 weakens from its *quantitative* form (a live-width bound *computable* from  $C_0$ —the target behind most of the width-accounting defects the reviews found) to the *qualitative* core just isolated, and the computable bound then returns for free *a posteriori*.

The dovetailing reduction is itself machine-checked (`formal/SemiDecidability.lean`, zero sorry), and the FO transcription it rests on—that the axioms above carve out exactly the Lean frame conditions, and that the standard translation matches the certified satisfaction relation—is cross-checked in `wp84` (exhaustively for domains up to three points). Two honest caveats: the qualitative core of F6 is *untouched*, so this reshapes the target rather than hitting it; and the  $\Pi_1^0$  placement adds one standard external theorem (Gödel completeness) to those the argument already invokes (the RCC5 patchwork property; parity-automaton emptiness in the historical automata route). No complexity bound is claimed or implied.

## 7.7 The regular-cover pivot

**Idea.** A reframing suggested by Wessel: a finite *quotient* of a PP-chain is invalid (a PP-cycle contradicts EQ-identity), but a finite *generator* is not—a state with a PP-self-loop *unfolds* to an infinite chain of distinct copies. So replace finite quotients by *regular covers*: finite presentations whose infinite *unfoldings* are the models, in the spirit of the two-tier quotient but for the full logic. This finer invariant can represent exactly the patterns that fool a width bound—identity selectors ( $i = j$  via address equality), half-graphs ( $i < j$  via an order test)—finitely.

**What it produced.** A complete *local* algebra of strong-EQ RCC5 networks, one piece of which is machine-certified:



**Theorem 15** (RCC5 normal form). *A strong-EQ atomic RCC5 network is composition-closed if and only if it is an ordered-disjoint structure: PP is a strict partial order, DR is a symmetric, irreflexive, downward-closed disjointness (if  $x \text{ DR } y$  and  $x' \leq x$ ,  $y' \leq y$  then  $x' \text{ DR } y'$ ), comparable pairs are never disjoint, and PO is the residual. The forward direction (every RCC5 network is ordered-disjoint) is machine-certified for arbitrary domains (`formal/RCC5NormalForm.lean`, axioms `propext` only). The converse is now certified too: GPT-5.6 Pro’s technical note [42] (`papers/final_paper_gpt-5.6_review/ALCI_RCC5_F6_technical_note.pdf`) gave an explicit canonical set representation—each element  $x$  maps to the non-disjoint pairs with a coordinate  $\leq x$ —and its core is machine-checked for arbitrary domains (`OrderedDisjoint.sub_iff_le`, `eta_injective`: zero axioms; `disj_iff_eta_disjoint`: + `Classical.choice` for the one non-constructive direction), independently cross-checked past size four in `wp88`. So the biconditional holds on arbitrary domains.*

With it come free amalgamation of ordered-disjoint structures, an exact one-point extension criterion, and exact uniform/non-uniform cross-policy characterizations—together *exhausting* the local algebra (`rcc5_local_amalgamation_theory`, `regular_cover_route_pivot/`, probes `wp45–wp83`). The route independently re-derives Theorem 10.

**Why it is a better map, not a shorter path.** Its own keystone (a bounded coherent regular-cover property) is F6 relabelled, and the route concedes, after exhausting the local algebra, that the remainder “cannot be a purely local RCC5 composition problem.” Because RCC5 relations are total, every model’s Gaifman graph is complete, so a one-dimensional PP-chain already has unbounded clique number and treewidth; no off-the-shelf guarded or bounded-treewidth cover theorem applies, and the keystone would need a bespoke cover theorem for complete edge-coloured regular structures. The pivot is the right vocabulary for a future attack—but it does not move the summit.

## 8 Current implementation: reasoners

The project ships working reasoners, on the same foundation as the proof attempts but validated empirically rather than proved.

**The cover-tree tableau.** `src/cover_tree_tableau.py` ( $\approx 350$  lines) is an operational concept-satisfiability procedure with exact-occurrence blocking, resting on the split-forest normal form and patchwork completion (§7.4). It is cross-validated against an independent quasimodel oracle on 911 concepts (zero mismatches after the known blind spot is excluded) and against an independent model verifier (`src/model_verifier.py`). SAT outputs accompanied by an explicit finite model are independently checkable by that verifier; no completeness theorem is claimed, and UNSAT outputs and SAT cases requiring infinite models remain empirical. Its one known limitation is the strong-EQ PO-loop, decided correctly by the tableau but not by the quasimodel oracle. It is strong evidence, not a proof.

**Evidence statistics, and the absence of a counterexample.** The empirical case is substantial, and it is worth stating precisely because it is the strongest thing short of a proof. The two reasoners agree on a 911-concept cross-validation suite (zero mismatches after the single known PO-loop blind spot of the quasimodel oracle is excluded), and the cover-tree decomposition passes 775/775 of its structured tests. We state its scope honestly: the two implementations are genuinely different algorithms (tableau versus type elimination) but share the composition table and test harness, so this is cross-validation with some common-mode risk, not two wholly independent oracles; and the model verifier confirms explicit *finite* SAT witnesses, not UNSAT verdicts

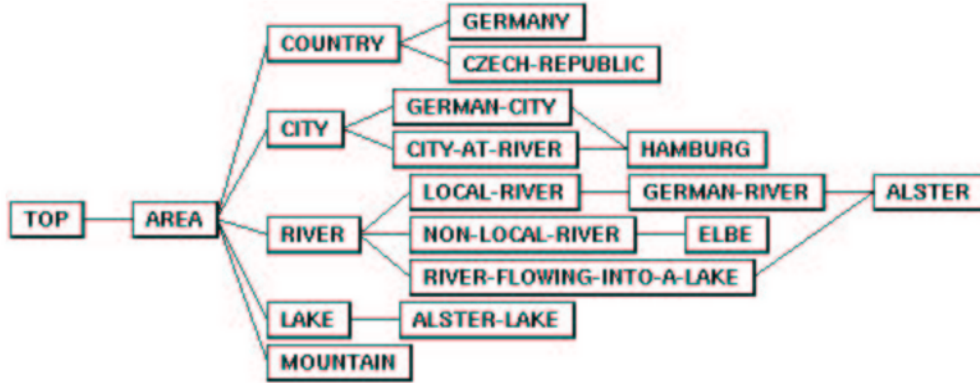


Figure 4: The GIS concept taxonomy *computed by a prototype reasoner* in Wessel’s 2002/2003 report (report7, Figure 6): the original (“the subsumption hierarchy ... was actually computed by a working prototype system,” report7, §4.3.1).

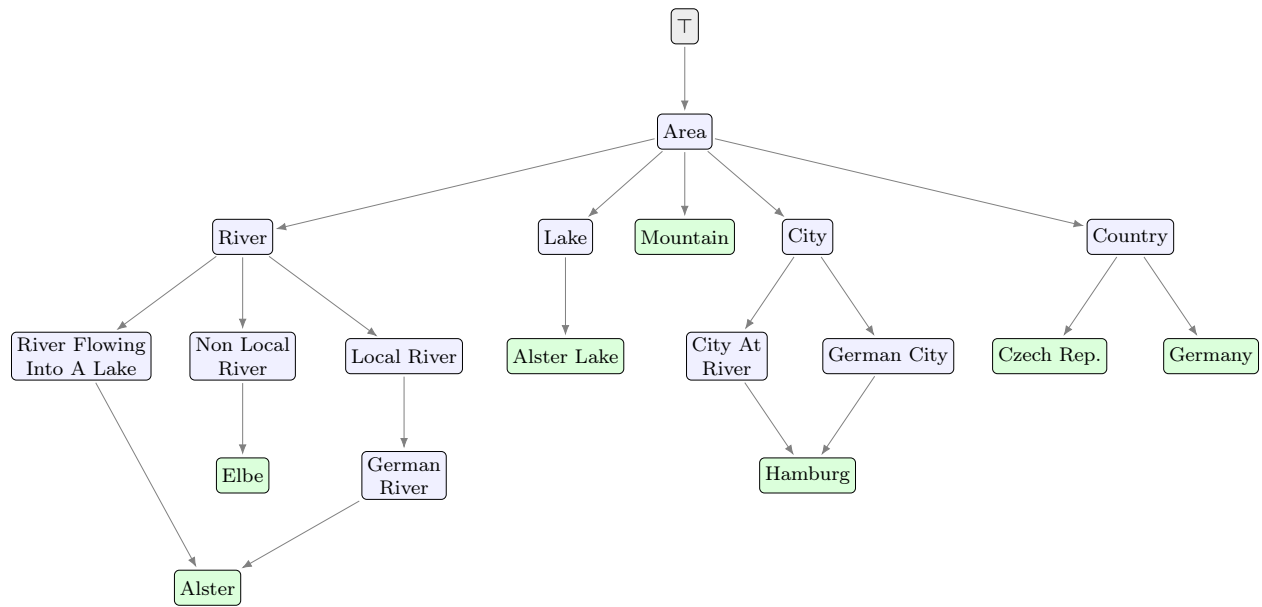


Figure 5: The *same* GIS taxonomy recomputed in 2026 by the cover-tree tableau reasoner (`src/gis_taxonomy.py`), rendered directly from the reasoner’s own output (arrows point from the more general concept to the more specific; green = individuals). All 21/21 subsumptions match Figure 4 edge for edge. Note that *Hamburg* (a German city at a river) and *Alster* (a German river flowing into a lake) each have two parents: the taxonomy is a directed acyclic graph, not a tree—exactly the multiple-inheritance the reasoner must get right.

or infinite-model SAT. Beyond the reasoners, some eighty self-contained verification work packages (`verification/python/`, WP1–WP88) cross-check the finite combinatorial content of every proof route against the RCC5 composition table *re-derived from finite set semantics*, so the underlying algebra cannot be fudged. Representative results: the RCC5 patchwork property holds on 63,219 two-bag amalgamations plus 300 tree-shaped patchings with zero failures (WP14); the finite acceptance mechanism matches the cover-tree tableau on ten diagnostic concepts plus a 148-concept random sweep with zero disagreements (WP16); the cluster-quasimodel decision layer matches the tableau on 11/11 diagnostics plus a 200-concept sweep (WP26); the RCC5 normal form agrees with brute-force composition closure on all networks up to size four (WP47); and the Lean development builds with zero `sorry` on every run (WP34). Most tellingly, *across all thirty rounds and seventeen adversarial reviews, no accepted-but-unsatisfiable concept was ever found*: every defect the reviews turned up was a gap in an *argument*, never a semantic counterexample to the intended theorem (the absence of a hybrid  $\downarrow$ -binder—a way to name the current point and reuse that name inside nested modalities—appears to block concept-level exploitation of the seams the reviews located; note that *nominals* themselves *are* definable under strong EQ, so it is the binder, not nominals, that is missing). This is not a proof—no amount of it can be—but it is the reason we describe the decidability argument as *strongly supported*.

**A GIS example spanning 23 years.** The tableau recomputes the GIS concept taxonomy of Wessel’s original report [9] (its Figure 6, the 2003 original, reproduced here as Figure 4): all 21/21 subsumptions are recovered (`src/gis_taxonomy.py`), and the reasoner emits its own rendering of the resulting hierarchy, shown as Figure 5. A taxonomy computed in 2003 by a *prototype reasoner* of unproven soundness (report7’s own caveat) is reproduced, in 2026, by an AI-written reasoner resting on the split-forest foundation—two reasoners, twenty-three years apart, agreeing edge for edge.

**How the tableau derives it.** The GermanCity subsumption of the motivating example (§1.2) is a good window on the calculus. To decide  $\text{Hamburg} \sqsubseteq \text{GermanCity}$  the tableau tests  $\text{Hamburg} \sqcap \neg \text{GermanCity}$  for *unsatisfiability*. Unfolding,  $\neg \text{GermanCity} = \neg \text{City} \sqcup \exists \text{PP} . (\text{Country} \sqcap \neg \text{Germany})$ ; the  $\neg \text{City}$  disjunct clashes at once (Hamburg is a city), so the procedure must realize a PP-successor  $c$ —a country other than Germany—beside Hamburg’s asserted PO-successor, the Alster. Here the *cover-tree* step does what the hand argument did: closing the node under composition forces the Alster– $c$  edge, and the Alster’s  $\forall \text{PO} . \neg \text{Country}$  excludes the PO option of  $\text{comp}(\text{PO}, \text{PP})$ , so the edge is PP. The Alster’s  $\forall \text{PP} . (\neg \text{Country} \sqcup \text{Germany})$  then propagates to  $c$ , forcing  $c$  to be  $\neg \text{Country}$  or  $\text{Germany}$ —contradicting “country other than Germany.” Every branch clashes, the concept is unsatisfiable, and the subsumption holds. The two ingredients—*composition-closure* of the interface and *universal propagation* along the forced edge—are exactly the operations the split-forest procedure performs at every node.

## 9 On agentic AI usage

We report candidly on the workflow, because the methodology may outlast the mathematics.

**The loop.** Wessel posed the problem, supplied the central intuitions and directions (the split-forest decomposition, the cluster/loop and regular-cover reframings, the honesty discipline), and after each round reviewed, corrected, and re-prompted. An AI proposed a proof or a repair; a *fresh* AI session—with no memory of how the proof was built—then reviewed it adversarially. Repeat.

**Division of labour, and a calibrated comparison.** The roles were asymmetric: one model (GPT-5.5) chiefly authored proofs, another (Claude) chiefly implemented, verified, audited, and documented, and the human supplied the direction and the original insight. It is tempting to read the review record—each model repeatedly finding defects in the other’s drafts—as ranking them, and we resist that inference: it is confounded by role (proof-authoring and cold-reviewing were never swapped), by model version (both lineages advanced mid-project), and by task (cold-reviewing a finished manuscript is a different, arguably easier job than authoring one). The robust finding, independent of which model is sharper, is that *neither is self-sufficient*: both write fluent long-context formal mathematics, both fail to reliably self-review, both at times declared results “closed” prematurely, and neither produced the decisive original idea. What carried the project was the loop, not any single model.

**An auditable record, for honest attribution.** Because “who contributed what” is exactly the question that AI-assisted work makes fraught, we kept the entire, timestamped conversation log public [64], alongside the round-by-round manuscripts and a superseded-work archive (`OUTDATED.md`). A reader can trace each idea, defect, and repair to its origin—human or model—rather than take our word for the division of labour above. We treat this as part of the methodology, not an afterthought: a human director should not quietly absorb credit for a model’s mathematics, nor should a model’s fluent prose obscure the human’s insight, and the only reliable guard is a record open to inspection.

**Cold review as a standard step.** The single most transferable finding: a *cold* reviewer consistently found defects that a *warm* reviewer (primed by the construction) missed—including, on this project, unsoundness proofs and scope errors that two hot audits had passed over. Across twenty reviews, a defect or overclaim was found in all but two. We recommend cold, cross-lineage adversarial review as a default intermediate step in AI-assisted mathematics, not an optional one.

**Don’t trust, verify.** Every load-bearing claim was cross-checked against something mechanical: self-contained Python probes that re-derive the RCC5 composition table from finite set semantics (so the algebra cannot be fudged), and ultimately the Lean kernel. Overclaiming was treated as the cardinal sin—and it happened: the author model twice overstated what the Lean established, and a cold review caught both; the claims were withdrawn. A sixteenth (cold) review of this very manuscript then found a *definite* error—an illustrative one-point-extension example whose claimed contradiction does not hold, the two triangles in fact forcing the single relation PO by *joint* constraint (Remark 2)—and, with it, that the informal “PO is never forced” intuition had been stated too strongly. Both were corrected, and the distinction between single-step and joint forcing is now made explicit throughout. That an error of this kind survived into a tagged manuscript is exactly why the standing label is *not certified*: the machine checks, not the model’s (or the author’s) confidence, are the ground truth. A seventeenth review—cold, scope-aware, and told

explicitly that this is an overview—then confirmed the joint-forcing correction had landed and found no new counterexample to the normal form, the conditional soundness theorem, or the  $\forall\text{PO}$ -free result, but flagged a residual cluster of *calibration* slips: an identity-selector minor was said to *prove* undecidability (it refutes only *this* width architecture); “completeness of our checker” was conflated with decidability itself (it is sufficient, not equivalent); and a blanket “forced merges are only self-defeating” contradicted the paper’s own satisfiable one-level gadget. All were corrected in the present version, and the review’s recommendation was to post the overview to arXiv as a status report and research handoff—not as a claimed solution.

**What frontier models did well, and did not.** Both models sustained long-context, rigorous, multi-round formal development and produced genuinely useful adversarial reviews and machine-checkable artifacts. What was *not* demonstrated is the decisive novel idea—no model cracked F6, and the honest expectation is that none will by the incremental means used here. The models excel at rigorous local formalization and at finding flaws; the genuinely new keystone is where human insight (or a qualitatively different search) is still required. We regard the AI contributions as co-authorship-grade in a human collaborator; we also regard the human’s role—problem, intuition, direction, and the discipline of honest labelling—as indispensable.

**A failure mode worth naming: productive-looking thrashing.** The sharpest negative finding is not an error but a *pattern* of errors the model could not see. Over roughly three weeks the assistant proposed four successive local disciplines to repair one architecture — a finite node set reused by blocking and witness borrowing. Each proposal was locally reasonable, each was implemented and probed, and each fell to an exact finite countermodel. The assistant then proposed the next one. What it did not do was conclude that the *class* was wrong, and this was not for want of the relevant fact: its own design document already recorded that this was the same pointwise-versus-joint shape that had broken three earlier architectures, and noted it as the fifth occurrence. The record was there and went unapplied.

We think this is a characteristic hazard of long-context formal work rather than an incidental lapse. Each repair is a genuine advance on the last, the countermodels feel like information about *this* attempt rather than about the family, and the accumulated context makes the surrounding framing feel settled precisely because it has never been questioned in that context. It is not hallucination and not local error; every step was correct and checked.

What broke it was cheap and came from outside the mathematics: the human observed that the loop was thrashing, called a halt, and asked a *different model* — GPT-5.6 Sol, with no history in the architecture — for an alternative certification route rather than another repair. The reply retired the assumption the whole architecture rested on (node reuse) and became the route that certified the fragment. The instrument is the same one as cold review, pointed at *design* instead of verification, and the lesson generalizes past this project: when successive fixes to one design keep failing, the cheapest useful move is a model that has never seen the design. The judgement that the moment had arrived was the human’s, and we do not know how to automate it.

**A worked example: a seed became a fragment.** The clearest single piece of evidence for the value of the loop is not the F6 work but the  $\forall\text{PO}$ -free fragment (§§7.2–7.3), and it is worth being precise about the *shape* of that contribution. Wessel supplied a seed whose consequences he could not compute in advance: collapse same-type occurrences on a PP-chain into a kernel node with a reflexive PP-loop. The models then did the part that is tedious and error-prone by hand—the exhaustive algebra (reflexive PP is self-absorbing; PP-cycles are impossible; single-type chains

collapse but multi-type ones do not), the two-tier construction that routes around the obstruction, and, the decisive step, locating exactly where the idea *breaks*. The PO gap was in neither the seed nor the first construction; it surfaced under cold review, and the result crystallized precisely at that boundary—the fragment is defined by exactly where the seed stops working. No single participant produced this: human intuition chose *which* idea to chase, the construction turned it into a real object, and the cold review caught the gap the construction had missed. That is the loop of this paper exhibited on a concrete result rather than asserted—and it is the honest, bounded form of the title question. Not that agentic AI settled the open problem, but that a human–model loop turned a vague structural hunch into a decidable fragment with a machine-checked *decision procedure* that *none* of the participants, human or model, would have produced alone. One caveat we kept in view for a long time was that this is a new result *of the collaboration*, which is a different question from whether it is new relative to the prior description-logic literature. That second check has since been made (§4.4), and its outcome is worth stating in the weak form the evidence supports: Lutz and Wolter leave  $L_{RCC5}(RS)$  explicitly open and float relation-dropping as a candidate route without attaching an argument to it; report7 canvasses restricted fragments, but strictly stronger and finite-model-property-based ones that could not reach this fragment; and we find no decidability argument for any PO-restricted RCC5 language there or since. We therefore say the result is new *so far as we can determine* — deliberately weaker than a priority claim, since one literature search is not a systematic citation sweep. The *process* contribution is unambiguous either way.

## 10 Summary and conclusion

**What we learned.** A twenty-five-year-old open problem has been *reduced*, honestly and largely machine-checkably, to a single well-posed lemma (F6, bounded live width): the same horizontal-coordination obstruction that has blocked every known attempt to prove the problem undecidable—a *shared* obstruction, not a proof that F6’s failure would by itself entail undecidability. The local algebra of RCC5 is exhausted and now certified. And one *sub*-question was settled outright rather than reduced: the  $\forall$ PO-free fragment is decidable, by a procedure the Lean kernel checks end to end, on raw input and under concrete set semantics, with fragment membership its only hypothesis (§7.3). Adversarial cold review plus machine verification is a repeatable methodology for this kind of work.

**What would move the needle.** F6 is a *definability* question—can a formula *force* unbounded rigid horizontal structure?—so the tools that fit are model-theoretic: locality (Gaifman/Hanf), Ehrenfeucht–Fraïssé / back-and-forth over bounded tuple profiles, the composition method, and bounded-model machinery. The concrete positive target is a *selector-sharing lemma*: that many same-profile selector gadgets must coalesce, so unbounded identity-selector minors cannot be forced. The concrete negative target *for static F6* is a concept forcing such minors; this would refute the bounded-live-width certificate architecture, but not, by itself, settle undecidability—an undecidability theorem would require an *additional* reduction showing how those minors encode a tiling or other undecidable computation. Ordinary locality alone is insufficient because horizontal degree is unbounded; this is the “correct room,” not a solved problem.

**The chances of closing F6.** We are frank: they are low, and re-treading the ground covered here will not change that. F6 is the crux on which this architecture turns—though, strictly, only  $F6 \Rightarrow$  decidability is established, and a *failure* of F6 would refute *this* certificate route without by itself proving undecidability (the logic could still be decidable by some representation this width



measure does not control). The two independent cold attacks converged on the same wall—evidence that the problem is correctly *mapped*, not that it is about to yield. It would be a poor use of effort to reconstruct the split-forest, automata, Lean, or regular-cover developments; the value of what remains is a genuinely new definability-theoretic idea, or a machine-assisted search of a different kind (below).

**The sharpened target: automatic covers, not static width.** The target has since shifted, and it is worth stating precisely, because it changes *where* to dig without changing the difficulty. Static F6 (a *computable* width bound) is now understood as *sufficient but not necessary*: identity selectors have unbounded static separator cover yet are word-automatic (GPT-5.6 Pro’s technical note [42]), so static F6 could *fail* with the logic still decidable. The recursion-theoretic *equivalent* of decidability is weaker than any particular checker: that *some* fixed sound, decidable, complete certificate relation accept exactly the satisfiable concepts (equivalently, by the  $\Pi_1^0$  observation of §7.6, that every satisfiable concept have *some* finite certificate). Completeness of *this* split-forest/live-width checker is a *sufficient* target, not known to be necessary; and the most promising sufficient route is now the *automatic-cover* target: *every satisfiable concept has a tree-automatic model*, which yields decidability by the same dovetailing and, unlike static F6, *admits* such selectors. This is a better map—a standard object (automatic structures) with mature theory, and the width-specific obstruction defused. It is not, however, a shorter climb: one still owes a *positive* existence statement over all satisfiable concepts, and the hard gap is unchanged—“the bare pattern is automatic” is not “every concept-forced model is uniformly automatic,” and totality (complete Gaifman graphs, wp78) still blocks every off-the-shelf cover theorem. So the honest estimate is unmoved: better-mapped, with standard tools and a fully certified base camp, but the summit is where it was—and the crispest target, qualitative F6, is decidability itself in other clothes.

**The wins.** Independent of F6, this project produced: (i) an unconditionally *decidable fragment*, the  $\forall$ PO-free concepts—and, since 2026-08-29, a *machine-certified decision procedure* for it (`decidableFSat`, `decidableSat_cone`, `decidableSetSat`), covering all four quadrants, taking raw input, sound and complete for concrete set semantics, and attacked by three independent cold reviews without a counterexample; (ii) a machine-certified RCC5 normal form; (iii) a machine-checked *conditional decidability theorem* ( $F6 \Rightarrow \text{decidable}$ , soundness certified, resting on the F6 oracle); (iv) a *working, empirically validated prototype reasoner*—the cover-tree tableau (§8), cross-validated against an independent quasimodel oracle on 911 concepts with *zero* mismatches, reproducing 23 years later the full GIS subsumption taxonomy of Wessel’s original report (21/21), and across the entire campaign never once producing a decision contradicted by any other oracle or by a hand analysis—*no counterexample was ever found*; (v) a sharp *map* of the problem with a named toolbox; and (vi) a candid *methodology* for adversarial AI-assisted mathematics.

Three of these are the project’s flagship artifacts, and they are complementary rather than ranked. The certified decision procedure for the fragment (item i) is the one result here that is *finished*: a theorem, not a reduction, with nothing left conditional. The rest of the Lean development (items ii–iii) is a *certified* surround for the open problem: the kernel guarantees the soundness pipeline and the abstraction’s faithfulness contain no gap, while F6 itself stays outside. The reasoner (item iv) is the *empirical* counterpart: a real, terminating implementation, run on thousands of concepts, that agrees with an independent implementation everywhere it has been tested and reproduces a two-decade-old published taxonomy. Neither proves decidability—but that a running procedure and a machine-checked proof both point the same way, and that no adversarial search over the whole campaign ever exhibited an accepted-but-unsatisfiable (or rejected-but-satisfiable)

concept, is the strongest *evidence* short of a certificate that the conjecture is true.

**Is this a scientific contribution?** We believe so, with the labelling kept honest: not a resolution of the open problem, but its reduction—checkably, and with a certified surround—to an irreducible core, plus one fragment settled outright with a kernel-checked decision procedure, one certified lemma, working tools, and an honest account of an AI-assisted attack and its dead ends. That is an accurate and, we think, respectable terminal state for a discussion piece on a problem open for over two decades.

**If nothing else: a demonstration, addressed to our peers.** Whatever the eventual fate of the mathematics, this report is at minimum an expressive demonstration of what frontier AI models could do in serious formal research at the time of writing (August 2026): sustain a months-long, adversarially-reviewed attack on a twenty-year-old open problem, author and repair proofs across some thirty rounds, catch one another’s defects with machine-checked witnesses, drive the surviving core into a kernel-checked Lean development, and ship working, validated reasoners along the way. Every description-logic researcher—indeed every logician and mathematician doing serious work—should be aware that this is what the instruments can now do. Ignoring these capabilities does not advance science, and declining to use them is, we think, simply unwise. Informing our peers of exactly this was itself a major purpose of the present report—a message first offered through the community’s own venue and, as Appendix A documents, rejected on grounds that engaged neither the mathematics nor the message. We record it here again, in the open.

**Outlook: the human is the bottleneck.** Although the key insights, intuitions, and directional choices came from the human, each round still required the human to orchestrate, review, and rewrite the prompts—and at that cadence roughly eighty rounds consumed about four months. Autonomous sub-agents *directing* the research could plausibly automate the diminishing-delta local work (formalization, local lemmas, probe-driven counterexample search) and remove that bottleneck. The cost may be prohibitive and it sacrifices direct oversight, but it is worth attempting for one reason this project is unusually well-suited to it: the objective is clearly defined and *machine-verifiable in Lean*. A closed, adversarial loop—propose, formalize, let the kernel accept or reject—has a built-in ground-truth oracle, exactly the guard the manual cold-review methodology provided. The honest caveat: Lean verifies *proofs*, not the *truth of a conjecture*, so such a loop can tighten the certified surround but the summit (F6) still needs a genuine idea; orchestration alone is unlikely to supply it.

**Materials.** The machine-checked development is `formal/Round19Transport.lean`, `formal/RCC5NormalForm.lean`, `formal/ForcingReduction.lean`, `formal/SemiDecidability.lean`, and `formal/POFreeLift.lean` (see `LEAN.md`); the status report and the F6/W2’ analysis are in `papers/fable5_width_barrier/`; the decidable fragment in `papers/two_tier_quotient_ALCIRCC5` and `papers/po_free_fragment_ALCIRCC5`; the certified local theory in `papers/rcc5_local_amalgamation_theory`; the reasoners under `src/`; the probes under `verification/python/`; and the full audit trail in `CONVERSATION.md` and `OUTDATED.md`; and GPT-5.6 Pro’s canonical-representation note is `papers/final_paper_gpt_5.6_review/ALCI_RCC5_F6_technical_note.pdf` [42]. The repository is at <https://github.com/lambdamikel/alcircc5>. The state described in this paper is frozen at the git tag `certified-fragment-2026-08-29`; for reproducibility cite that tag (or its commit hash), not the mutable `master` branch. It supersedes `arxiv-candidate-2026-07-18`, which predates the  $\forall$ PO-free fragment’s certification (§7.3) and the three cold reviews of it. Filenames

below are relative to the repository root. Table 6 maps each principal claim to its verification level, its file or Lean theorem, and its probe.

| Claim                                                              | Status         | File / Lean theorem                                                                              | Probe        |
|--------------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------|--------------|
| Certificate soundness                                              | Lean-certified | Round19Transport.lean: finAccept_sound                                                           | wp34         |
| Hintikka faithfulness                                              | Lean-certified | Round19Transport.lean: satisfiable_iff_hintikkaP                                                 | —            |
| RCC5 normal form (both directions)                                 | Lean-certified | RCC5NormalForm.lean: toOrderedDisjoint, sub_iff_le, eta_injective                                | wp47, wp88   |
| $\forall$ PO-free lift core                                        | Lean-certified | P0FreeLift.lean: lift_cc, unf_is_frame                                                           | wp86         |
| F6 $\Rightarrow$ decidability (soundness)                          | Lean-certified | Round19Transport.lean: decidableSat_of_finScheme                                                 | —            |
| $\Pi_1^0$ / dovetailing reduction                                  | Lean-certified | SemiDecidability.lean: decidableSat                                                              | wp84         |
| $\forall$ PO-free <b>decidability</b> , NNF input                  | Lean-certified | P0FreeLift.lean: decidableSat_cone                                                               | wp133, wp134 |
| ...on <i>raw</i> input                                             | Lean-certified | P0FreeLift.lean: decidableFSat, nnfP_correct                                                     | —            |
| ...under <i>concrete set</i> semantics                             | Lean-certified | P0FreeLift.lean: decidableSetSat                                                                 | —            |
| Abstract = concrete satisfiability ( <i>full</i> logic)            | Lean-certified | P0FreeLift.lean: satisfiable_iff_set                                                             | —            |
| One-sided UNSAT test ( <i>full</i> logic)                          | Lean-certified | P0FreeLift.lean: coneScheme_unsat_full                                                           | wp134        |
| Completeness of the cone scheme                                    | Lean-certified | P0FreeLift.lean: coneScheme_complete                                                             | wp135        |
| The two <i>arguments</i> for the fragment ( $K(C_0)$ , extraction) | Theorem-level  | papers/po_free_fragment_ALCIRCC5; the certified route above is a <i>third</i> , and uses neither | wp87         |
| Identity-selector characterization                                 | Theorem-level  | papers/fable5_width_barrier                                                                      | wp44         |
| Cover-tree tableau                                                 | Empirical      | src/cover_tree_tableau.py (911-concept cross-validation)                                         | wp —         |
| Static F6; automatic-cover; full decidability                      | Open           | — (no proof claimed)                                                                             | —            |

Table 6: Result-to-artifact map. Statuses are those of Table 1. Every Lean name listed was checked to resolve in the named file at tag `certified-fragment-2026-08-29`; filenames are relative to the repository root. Note the split in the last two rows: the fragment *theorem* is certified, by a route that uses neither the  $K(C_0)$  bound nor the extraction argument, while those two *arguments* remain unformalized.

## Acknowledgments

The  $\mathcal{ALCZ}_{\text{RCC}}$  family was introduced by Wessel in his doctoral work at the University of Hamburg, under the DFG project “Description Logics and Spatial Reasoning” (grant NE 279/8-1); the author was additionally supported by a *Grundstipendium* of the University of Hamburg from 1999 to 2001. The present investigation was carried out by AI assistants—Claude Opus 4.6, 4.7 and 4.8, Claude Fable 5 and Claude Opus 5 (Anthropic), and GPT-5.4, GPT-5.4 Pro, GPT-5.5, GPT-5.6 Pro and GPT-5.6 Sol (OpenAI)—under the author’s direction, across many rounds of proposal, adversarial review, and revision; GPT-5.4 Pro contributed the split-forest semantics and cover-tree tableau formalization, GPT-5.5 the repaired decidability manuscripts and the majority of the formal proofs, and GPT-5.6 Pro the canonical set representation certifying the converse of the RCC5 normal form (Theorem 15) [42] together with the final scope-aware review of this overview. **GPT-5.6 Sol contributed the *ConeScheme* design [62]**, which is what carried the  $\forall\text{PO}$ -free fragment to a machine-certified decision procedure: after two rounds of cold attack had refuted the previous certification architecture—finite node reuse by blocking and witness borrowing—GPT-5.6 Sol was asked for an alternative and proposed retiring node reuse altogether, keeping the finite scheme only as a control graph and unfolding every non-EQ existential to a fresh occurrence. That single change turned the five obligations that had cost four refuted disciplines into structural facts. Claude carried out the Lean development. The  $\forall\text{PO}$ -free (PO-coherent) fragment of §7.2 has a specific lineage worth recording: the human author seeded it with the *PP-kernel* idea (collapse same-type occurrences on a PP-chain into a kernel node with a reflexive PP-loop); Claude built that into the two-tier quotient construction; and GPT-5.4 Pro’s review exposed the “PO gap,” which is what narrowed the result to the  $\forall\text{PO}$ -free fragment where it is sound. Per the discipline described in §9, the full record of who contributed what is public in `CONVERSATION.md` [64].

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## A On the DL 2026 rejection

An extended abstract of this work was submitted to the 2026 International Workshop on Description Logics (DL 2026) and rejected. We record the episode—and dissent from the decision—not merely out of grievance but because it bears directly on this paper’s subject: how a research community engages with an AI-assisted attack on one of its own long-standing open problems. The submission was labelled exactly as this paper is (*strongly supported, not certified*, an AI-assisted discussion piece), and its stated motivation, twice, was to raise awareness of the emerging capabilities of agentic AI for description-logic research. We contend it was rejected on the wrong basis.

1. **The technical objections targeted the optional appendix.** DL 2026’s own guidelines state that the appendix “will not be included in the proceedings” and is reviewed only at reviewer discretion. Yet nearly every concrete objection—missing definitions, a broken cross-reference, an under-specified lemma—concerned appendix material, not the extended abstract that is the reviewed artifact.
2. **The reviews attacked a position the paper did not take.** The abstract called the proof a “candidate,” wrote “if found correct by human experts,” and carried an explicit non-verification disclaimer. It was nonetheless assessed as a settled-theorem submission and rejected for not clearing that bar—a strawman.
3. **Honest hedging was quoted back as a fault.** The very disclaimers included to avoid over-claiming were cited as grounds for rejection. Punishing calibrated uncertainty inverts the incentive peer review should create.
4. **No review engaged the actual thesis** (agentic AI’s capabilities), nor the concrete contributions the submission *did* present: a working cover-tree tableau reasoner reproducing the 2003 GIS taxonomy, and an unconditionally decidable fragment. (The split-forest normal form, the reduction to a single keystone, and the machine-checked Lean soundness reported in the present paper came later, in the months *after* the rejection—so they cannot have been the basis for it.)
5. **The verdict was a value judgment, not a technical assessment.** The three reviews largely agreed on the defect inventory but split on the verdict—one reviewer believed the result true and named the right approach, yet recommended reject on presentation; another recommended accept. The decision was escalated to the Steering Committee, and the covering letter offered that the paper “could be a good scientific contribution if it . . . focus[ed] on the advantages and disadvantages of using LLMs . . . rather than claiming decidability results.” But that discussion—the advantages and disadvantages of agentic AI for this kind of research—is exactly what the submission’s main text *did* provide, alongside the concrete results above; it simply was not engaged with. The “acceptable” paper being described was, in substance, the paper already submitted. The objection was to the framing and the precedent, not the substance.
6. **A disclosure paradox.** The integrity choice—front-loading the AI co-authorship, in the title and from page one—was precisely what made the submission unacceptable, while the offered path to acceptance was to write it in standard single-voice style with a tail-of-paper note. A process that rewards understating AI’s contribution and penalizes overstating it selects for under-disclosure, not for science.



We grant one point: a single introductory sentence (“We hereby claim:  $\mathcal{ALCI}_{\text{RCC5}}$  is decidable”) read more assertively than the surrounding, carefully hedged text supported, and a constructive “tone this down and resubmit” would have been met immediately. That does not carry the verdict. There is, finally, a recursive irony: the rejected abstract had warned, by name, about the failure modes—network effects, not-invented-here bias, dismissive reviewing—that the rejection then illustrated. The deeper concern is not one abstract’s fate but that agentic AI now lets ideas reach decades-old open problems from outside the established networks, and a review process that answers this by rejecting on framing while ignoring substance constrains rather than protects science. Peer review exists to surface and stress-test new ideas, not to keep them out; when it functions as the latter, it has stopped doing the job the community delegated to it. The full extended abstract, the three reviews, and a detailed analysis are public ([d12026-rejected/](#); the open letter is [open\\_letter.md](#)); readers are encouraged to form their own view.